

Data Modeling and Machine Learning

Chapter 3 – Hands on Data Analytics for Everyone

October 12, 2026

北师香港浸会大学
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Contents

- **Introduction to Machine Learning**
  - Supervised and Unsupervised Learning
  - Classification and Regression
- Linear Regression (Supervised Learning)
  - Model
  - Performance Evaluation
- Classification (Supervised Learning)
  - How to Perform a Classification
  - Classification Tree Model
- Clustering Method (Unsupervised Learning)
  - Objective
  - Similarity Measures
  - Method 1: Hierarchical Clustering
  - Method 2: K-Means Method (Clustering by Partitioning)
- Lab (Demo): Unsupervised Learning

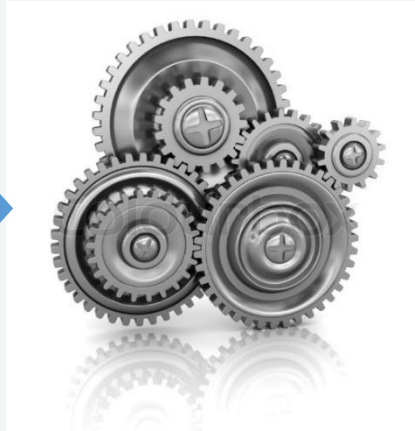
What is a Model (Learning Algorithm)?

A model or learning algorithm is simply a specification of a mathematical (or probabilistic) Relationship that exists between different variables.

- Input features
- Input attributes
- Independent variables

$$\boldsymbol{X} = (x_1, x_2, \ldots, xn)$$

Model



$$y$$

- Class
- Label
- Target
- Output feature/attribute
- Dependent variable

$$y = f(\boldsymbol{\beta}, \boldsymbol{X}) \quad \text{with } \boldsymbol{\beta} = [\beta_1, \beta_2, \ldots, \beta_m]$$

A learning algorithm adjusts (learns) the model parameters $\boldsymbol{\beta}$ throughout a number of iterations to maximize/minimize a likelihood/error function on y.

What Is Machine Learning?



Learning = Improving with **experience** at some **task**

Arthur Samuel (1959)

- Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.

Tom Mitchell (1998)

- Well-posed Learning Problem: A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

What Is Machine Learning?



Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam.

What is the task T, the experience E and the measure P in this setting?

1. Classifying emails as spam or not spam.
2. Watching you label emails as spam or not spam.
3. The number (or fraction) of emails correctly classified as spam/not spam.

Machine Learning Is Everywhere

Character recognition

- Raw data: image

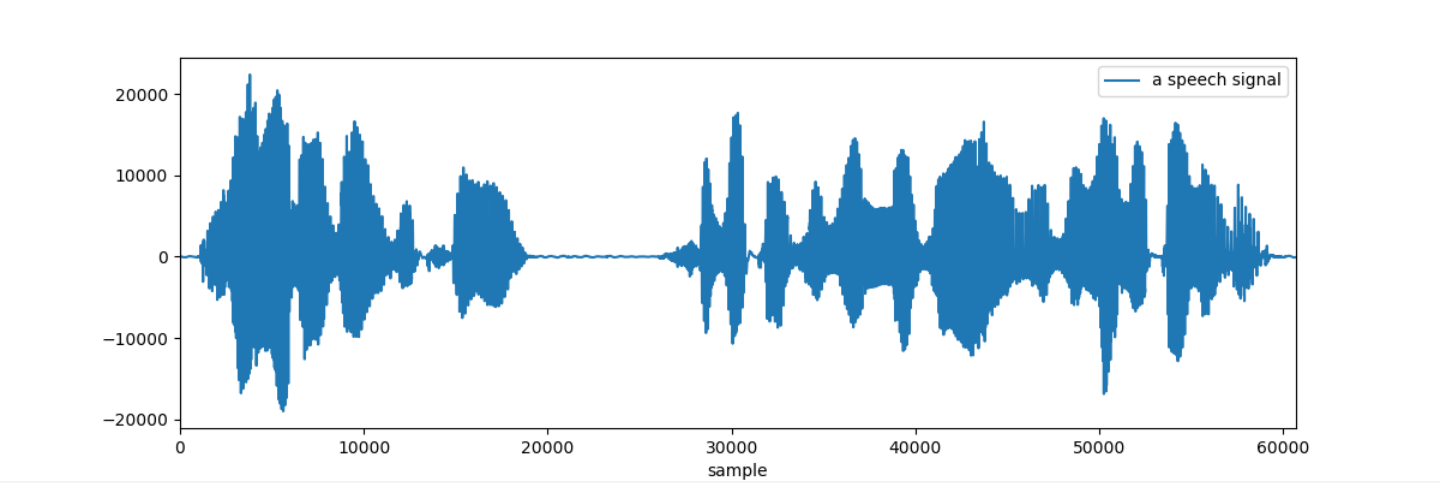


- Class: numerals, English (Chinese, etc.) characters

Machine Learning Is Everywhere

Speech recognition

- Raw data: speech signal



- Class: spoken words



Document Classification

- Raw data: (web) document

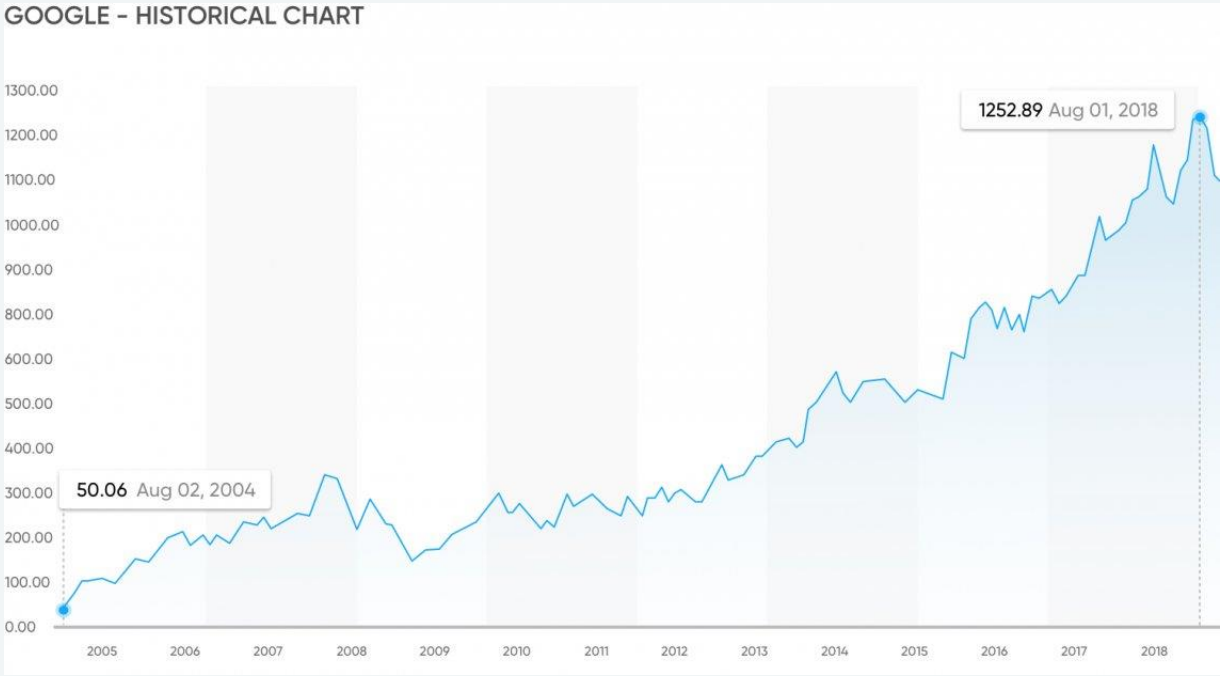
> *As the movie year winds down, I would like to express my gratitude to Martin Scorsese. Not only for making "The Irishman," his best movie in a long time and one of the best of 2019 (see below), but also for reminding the world of the value of cinema.*
> *The art form is in one of its periodic identity crises. A big chunk of our collective attention — we don't yet know how big, or with what consequences — is migrating to streaming platforms whose offerings include a lot of the stand-alone single-episode narratives that we used to see mainly in theaters.*

- Class: semantic categories (movie, art, money,...)

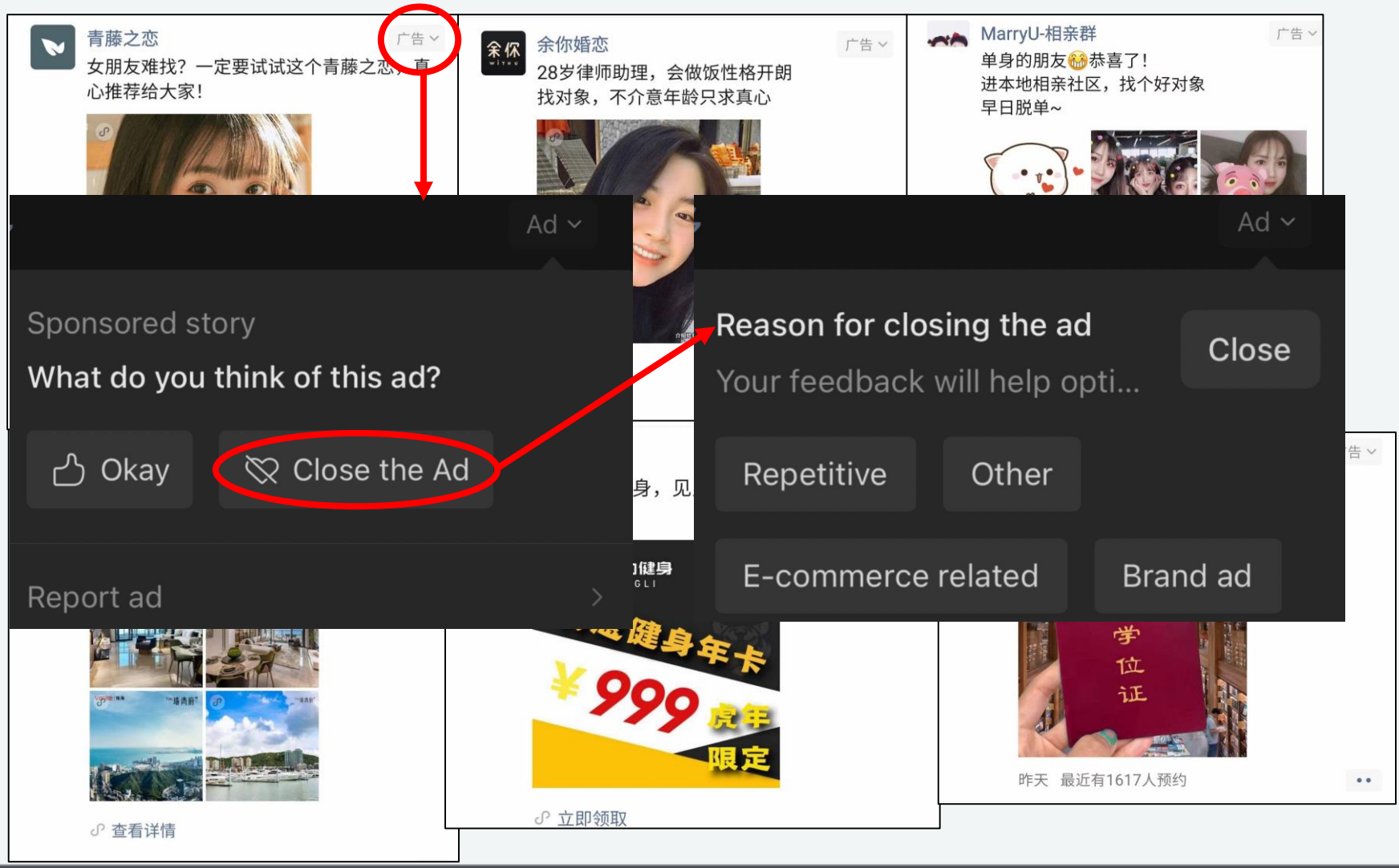
Machine Learning Is Everywhere

Financial Engineering

- Raw data: financial time series (e.g., stock prices)
- Task: classify financially healthy / unhealthy company, stock prediction, etc.



Machine Learning Is Everywhere



Machine Lear

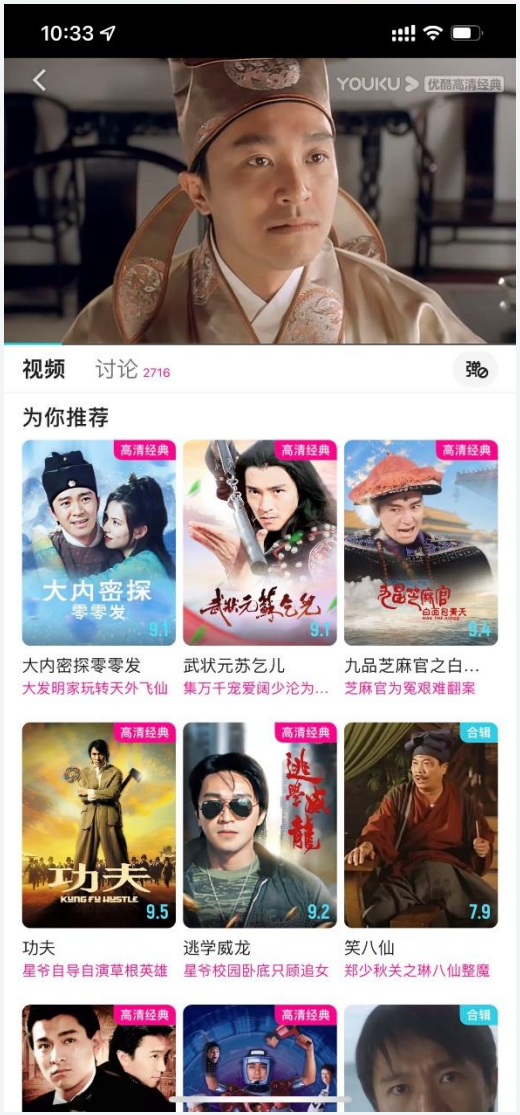
Many other exam

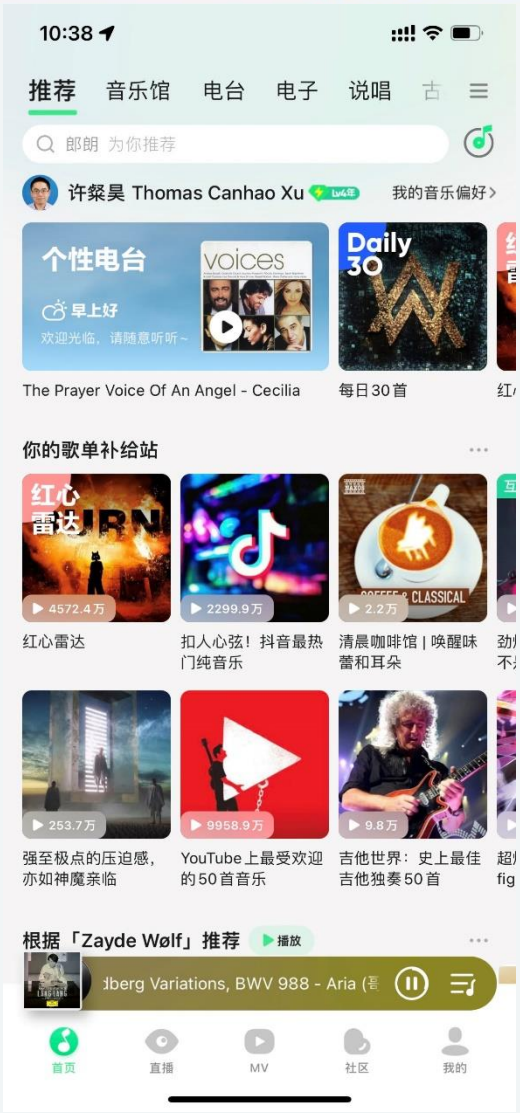
- Facebook: photo … your news feed
- Amazon: e-comm … casting demand, pricing
- NASA: identifyin … ers, galaxies, quasars, exoplanets, etc.
- Google Spreadsh … g to fill in missing values
- E-commerce: pre … clicked by users
- …



Recommendation with Machine Learning



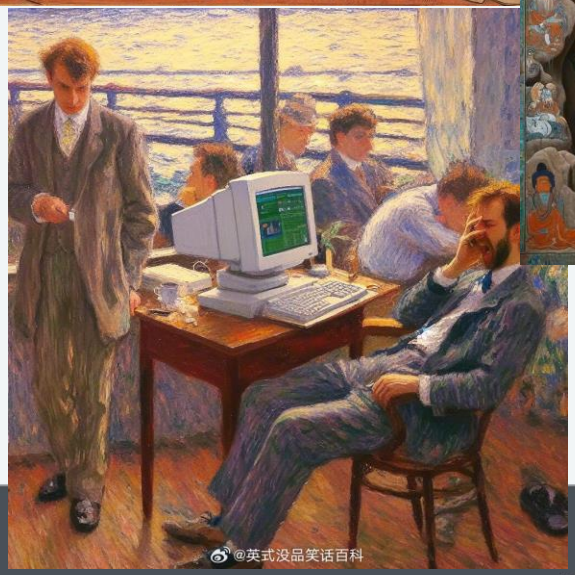




One of the hottest topic recently: AI generated art









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Learning Paradigm



- Supervised learning (有监督学习)
- Unsupervised learning (无监督学习)
- Semi-supervised learning (半监督学习)
- Reinforcement Learning (强化学习)
- Transfer Learning (迁移学习)
- Federated Learning (联邦学习)
- Feature Learning (表征学习)
- Edge Learning (边缘学习)
- Meta Learning (元学习)
- .....

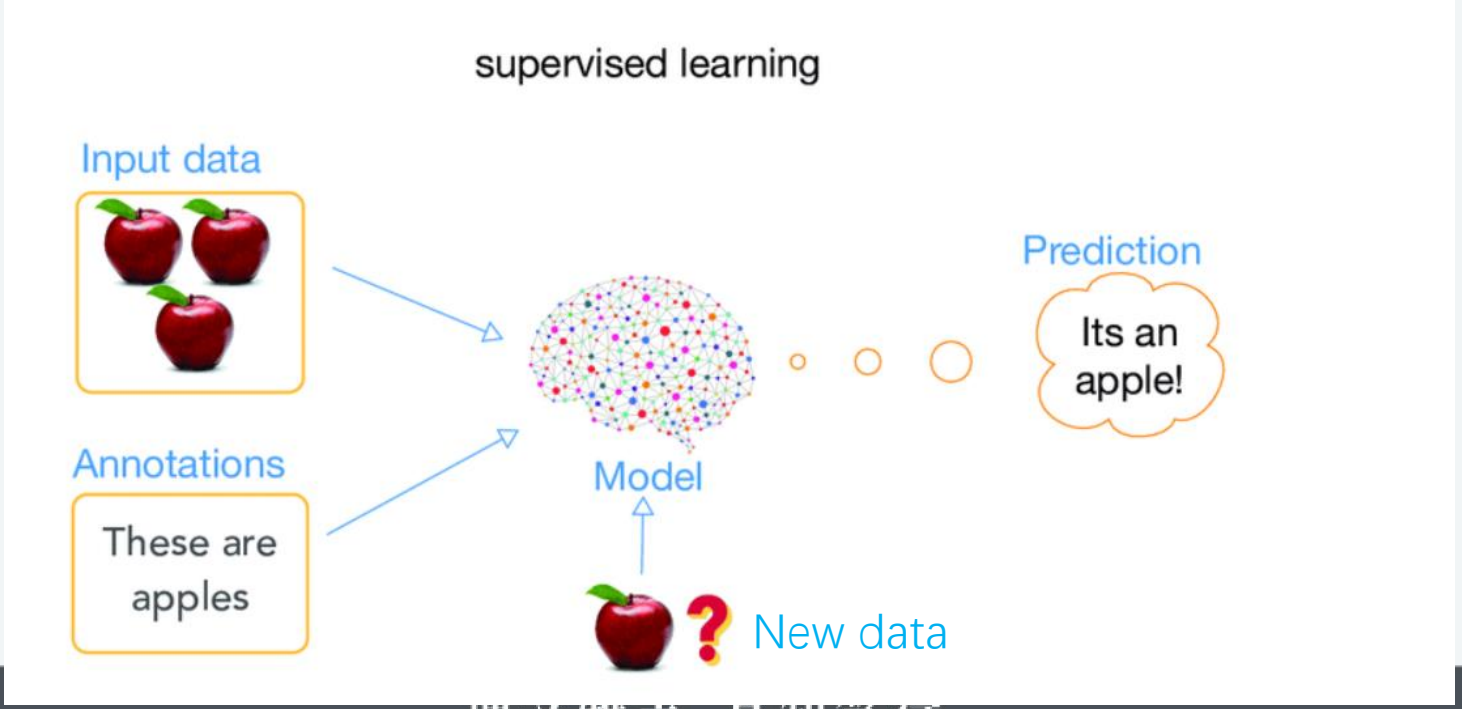
Supervised Learning

The learner is provided with **a set of data inputs together** with the corresponding **desired outputs**

- Data act as a “teacher”

Example:

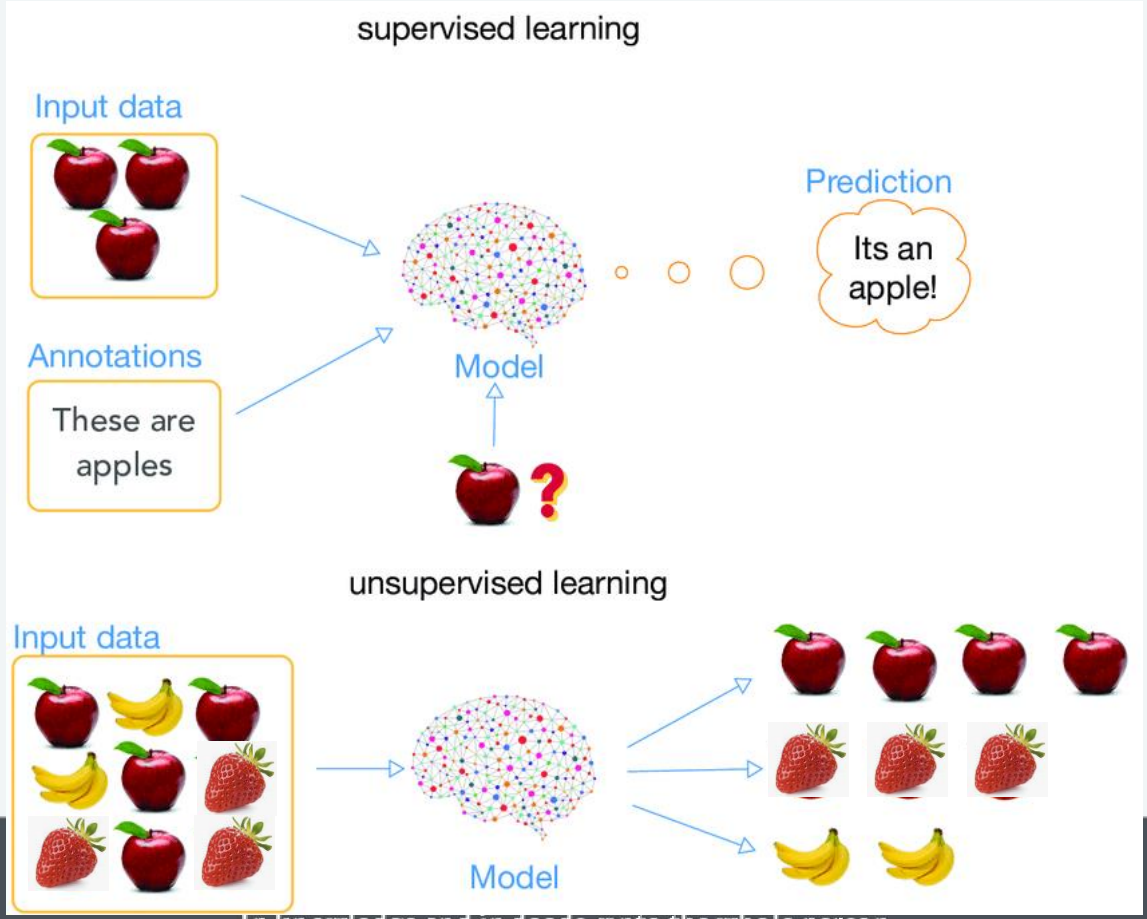
- Teach kids to recognize different animals



Unsupervised Learning

Training examples as input patterns, with no associated output

- no “teacher”
- **similarity measure** exists to detect groups/clusters



What's what?

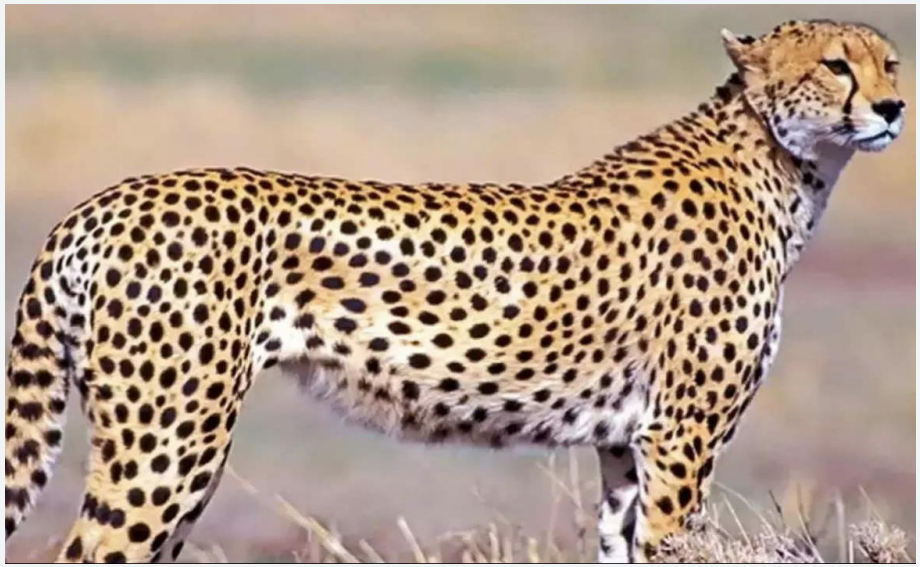


Assuming you are 3 years old

What's what?



What's what?









What's what?



What's what?



What's what?



What's what?


Cheetah
Snow Leopard
Clouded Leopard
Leopard
THE BIG CATS
Jaguar
Cougar
Tiger
Lion

What's what?



Supervised Learning Example



Example: Breast cancer (malignant/benign classification)

- Input: Tumor size samples
- Output: Whether the tumor is malignant or benign
- Task: Learn a classifier from the provided input and output, that can predict the label (category) for new tumor size inputs.

Supervised Learning Example



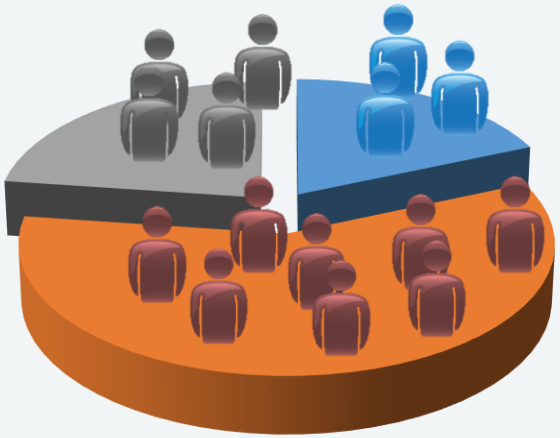
Example: House price prediction (Regression)

- Input: House size samples
- Output: House prices
- Task: Learn a model from the provided input and output, that can predict the house prices (quantity) for new house size inputs.

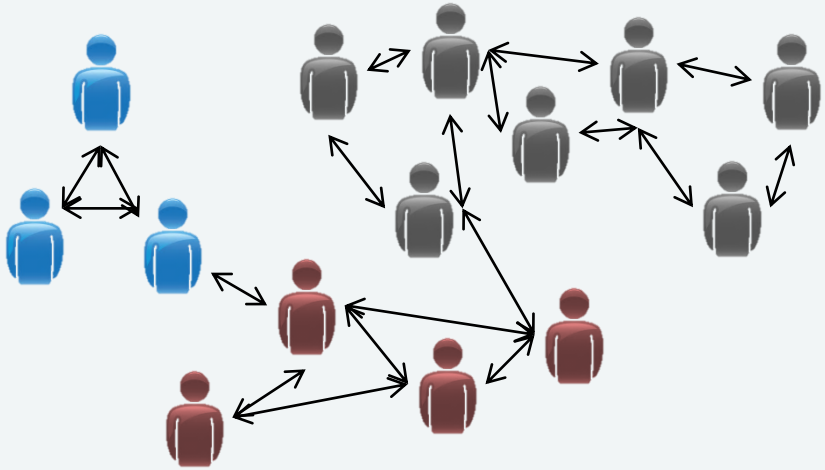
Unsupervised Learning Example

Clustering

In the early stages of an investigation, it may be helpful to perform exploratory data analysis to gain some insight into the nature or structure of the data



Market segmentation

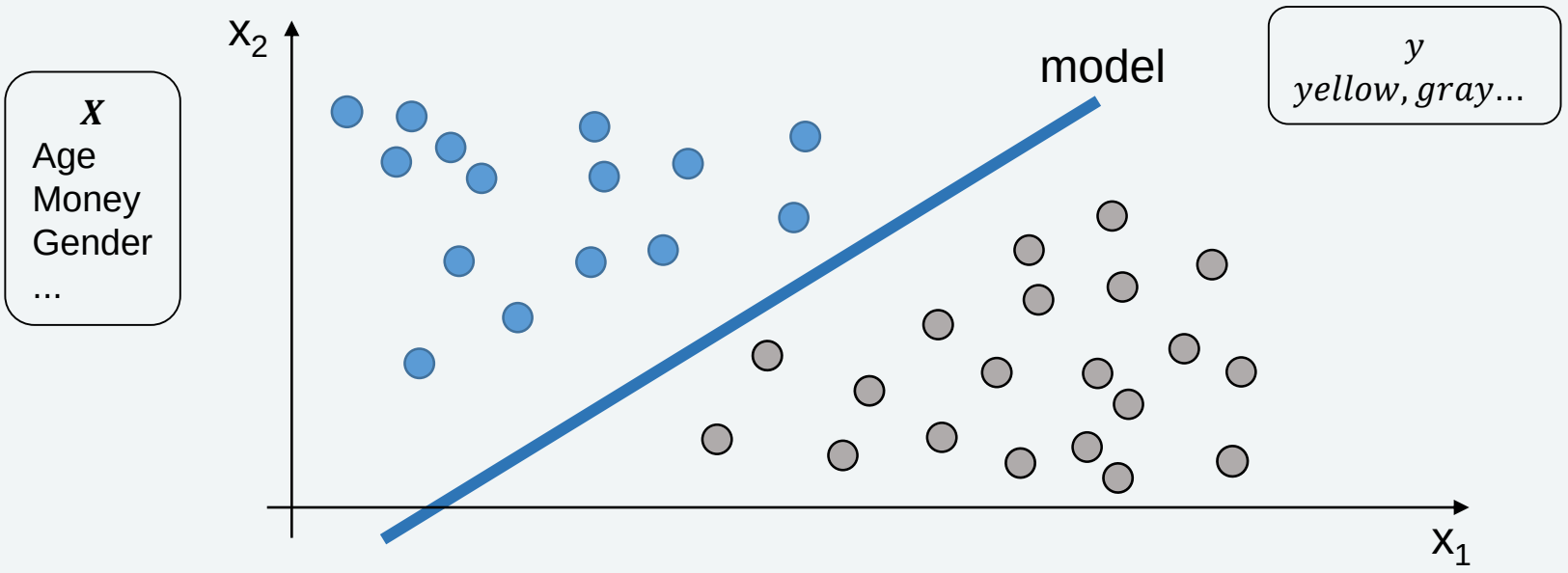


Social network analysis

Supervised Learning - Data Representation

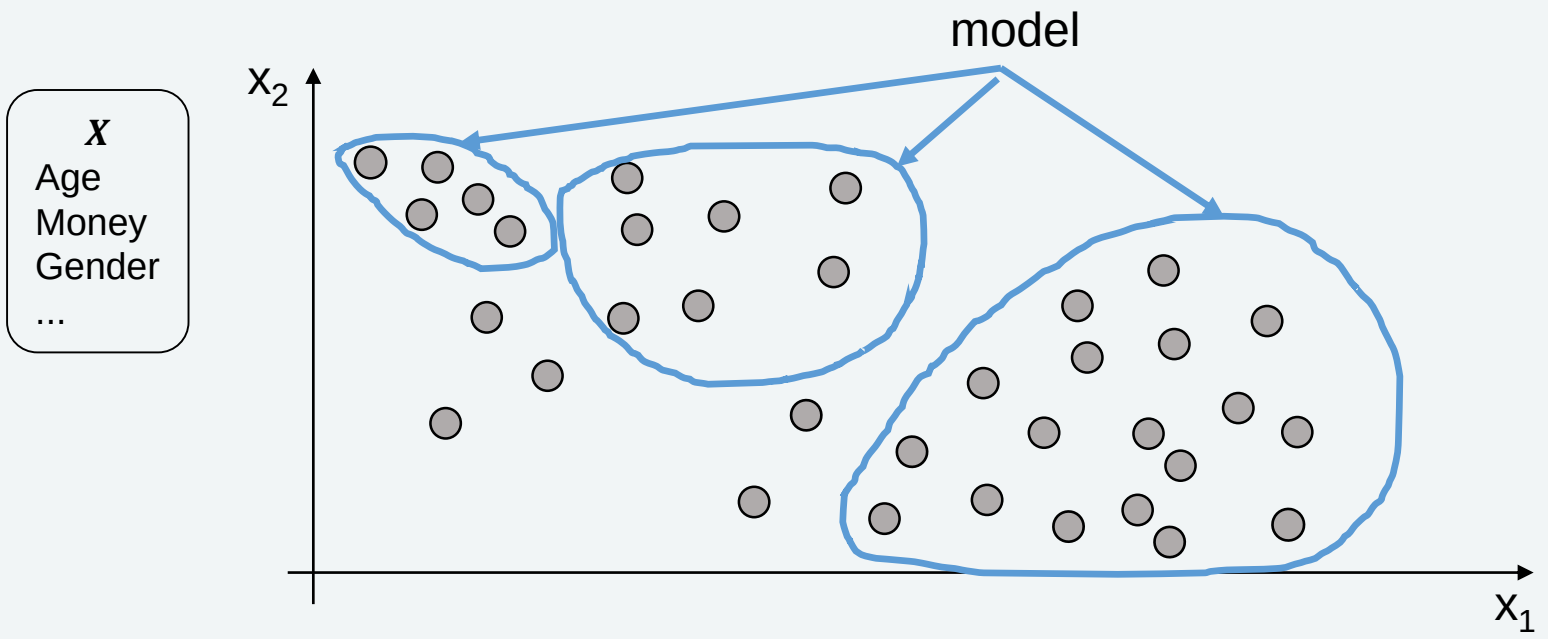
$\boldsymbol{X} = (x_1, x_2, x_3, \ldots)$ and $y = \{yellow, gray, \ldots\}$

- A training set with many examples of $(\boldsymbol{X}, y)$
- The model learns on the examples of the training set to produce the right value of y for an input vector $\boldsymbol{X}$



Unsupervised Learning - Data Representation

- $\boldsymbol{X} = (x_1, x_2)$ and ~~$y = \{yellow, gray\}$~~
- A training set with many examples of ($\boldsymbol{X}$, ~~y~~)
- The model learns to group the examples $\boldsymbol{X}$ of the training set based on similarity (closeness) or probability



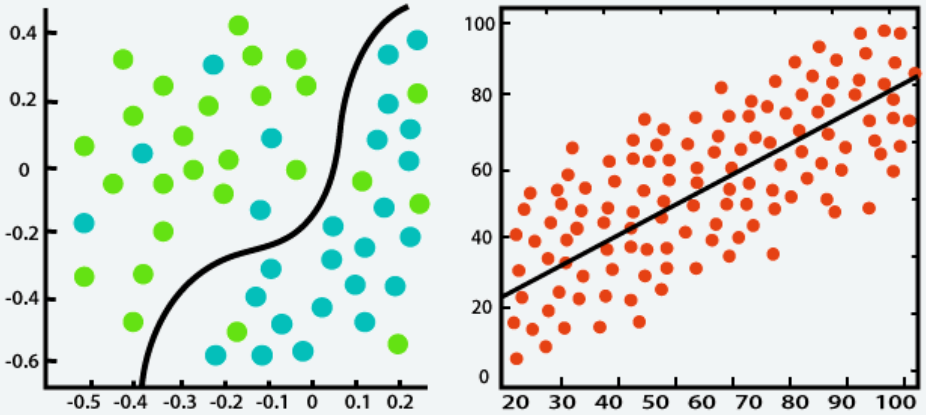
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Regression 回归 vs Classification 分类

Supervised learning: Given the “right answer” for each example in the data.

- When the target variable can take on only a small number of discrete values, we call it a **Classification** problem.
- When the target variable that we’re trying to predict is continuous, we call the learning problem a **Regression** problem.



Classification Regression

Classification

Given a collection of records: each record contains a set of attributes, one of the attributes is the class.

- Find a model (train a classifier) for class attribute as a function of the values of other attributes.

Previously unseen records should be assigned a class as accurately as possible.

- A test set is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

Classification Example



You are a bank loan officer and need to know which loan applicants are "safe" and which are "risky" for the bank

- Assign labels "safe" or "risky" to a loan applicant

You are a marketing manager at an electronics consumer shop and want to guess whether a customer will buy a new computer

- Predict the category of the customer, "buys" or "doesn't buy"

Classification is the task of assigning objects to several predefined categories/labels

Classification Example



	categorical	categorical	continuous	class
Tid	**Refund**	**Marital Status**	**Taxable Income**	**Cheat**
1	Yes	Single	125K	**No**
2	No	Married	100K	**No**
3	No	Single	70K	**No**
4	Yes	Married	120K	**No**
5	No	Divorced	95K	**Yes**
6	No	Married	60K	**No**
7	Yes	Divorced	220K	**No**
8	No	Single	85K	**Yes**
9	No	Married	75K	**No**
10	No	Single	90K	**Yes**

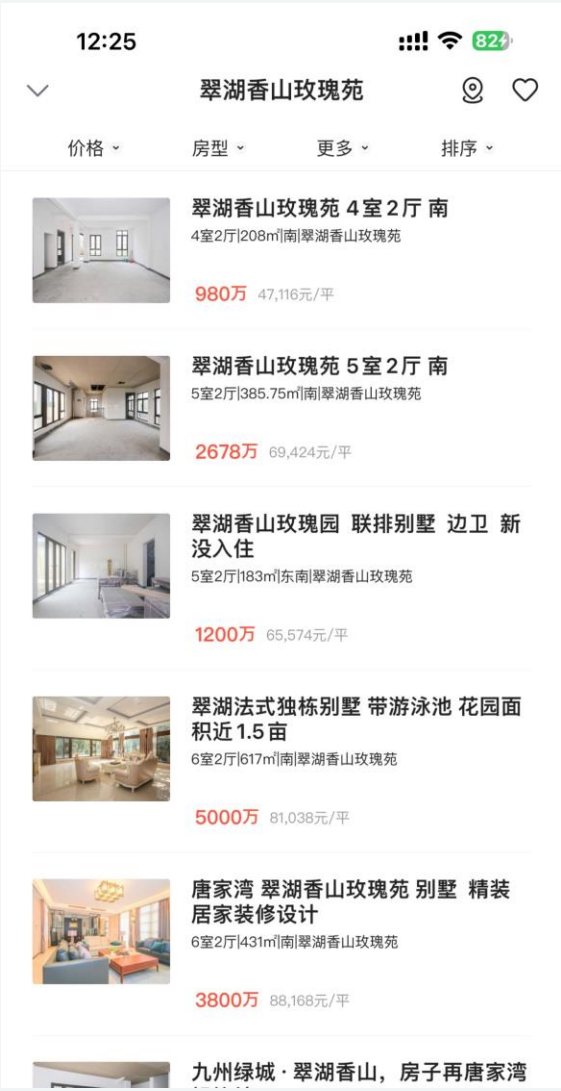
Tid	**Refund**	**Marital Status**	**Taxable Income**	**Cheat**
11	No	Married	80K	**?**
12	Yes	Single	100K	**?**

Regression Example





Suppose we have a dataset giving the living areas and prices of house nearby:

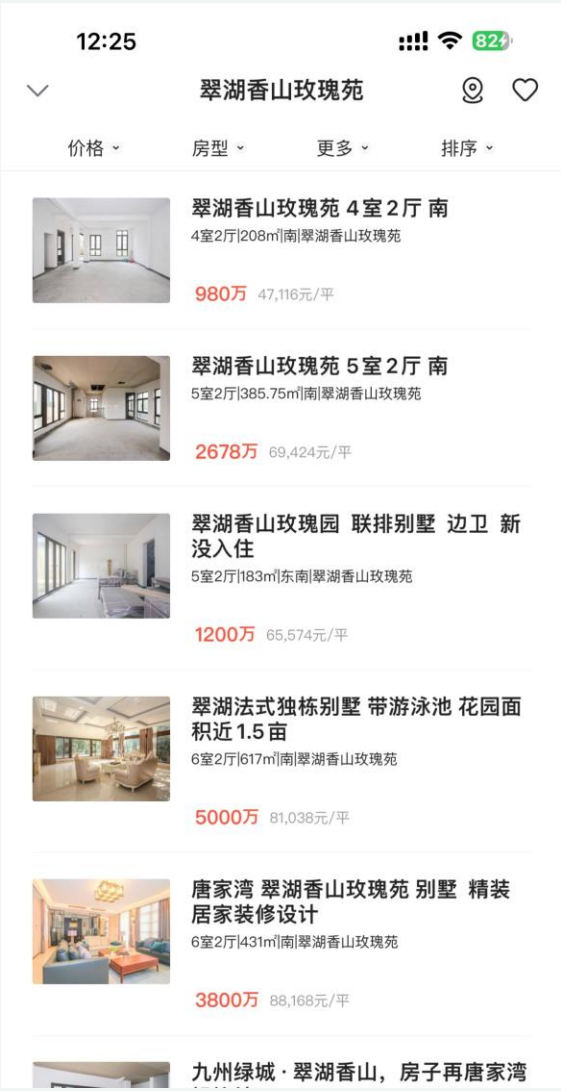


Regression Example





Can you predict the house price based on living area and number of rooms?



Supervised Learning Example



You're running a company, and you want to develop learning algorithms to address each of two problems.

- Problem 1: You have a large inventory of 4 similar items (14, 14 Plus, 14 Pro, 14 Pro Max). You want to predict how many of these items will sell over the next 3 months.
- Problem 2: You'd like software to examine individual bank customer accounts, and for each account decide if it has been hacked.

Should you treat these as classification or as regression problems?

Supervised Learning Example



For coronavirus, you want to develop learning algorithms to address each of two problems.

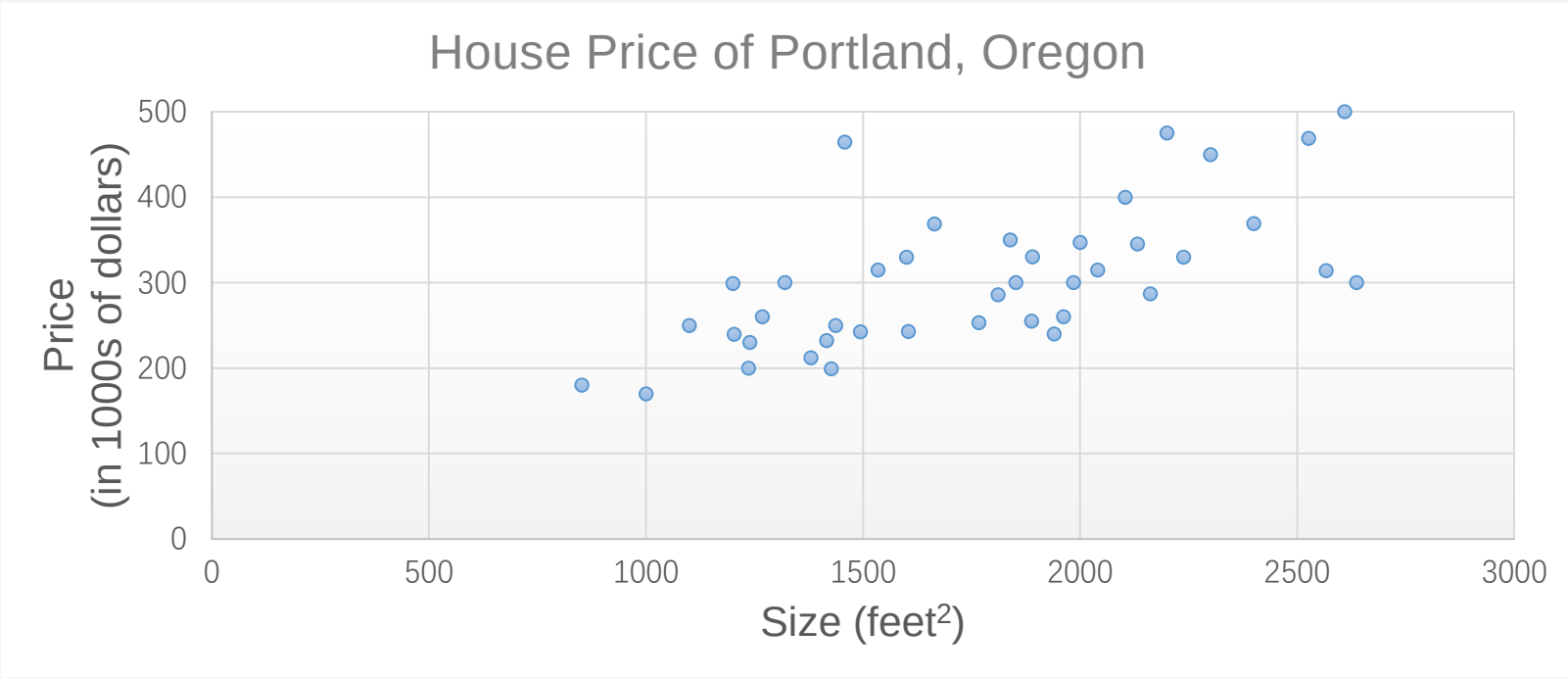
- Problem 1: You want to predict if someone is positive/negative of the coronavirus.
- Problem 2: You know today we have 5 coronavirus cases; you want to predict the number of cases for the next 7 days.

Should you treat these as classification or as regression problems?

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Regression Example



Given data like this, how can we learn to predict the prices of other houses in Portland, as a function of the size of their living areas?

Linear Regression – Problem Description

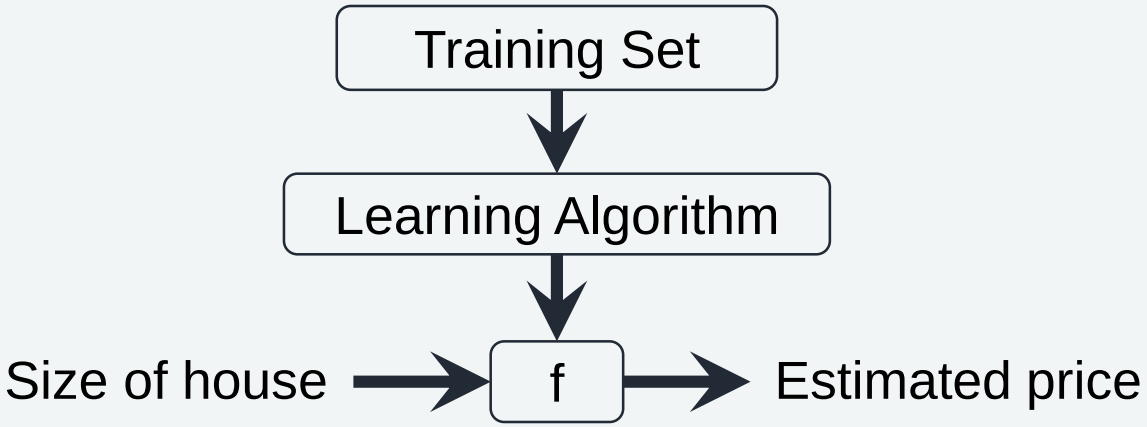
Notation:

- Input variable/feature: x
- Output/target variable: y
- Training example: $(x^{(i)}, y^{(i)})$
- Training set: $\{(x^{(i)}, y^{(i)}); i = 1,2, \dots, m\}$ (a list of m training examples)
- Space of input values: X ; space of output values: Y

To describe the problem slightly more formally, our goal is:

Given a training set, to learn a function (hypothesis/model) $f: X \longmapsto Y$, so that $f(x)$ is a "good" predictor for the corresponding value of y.

Linear Regression – Model Representation



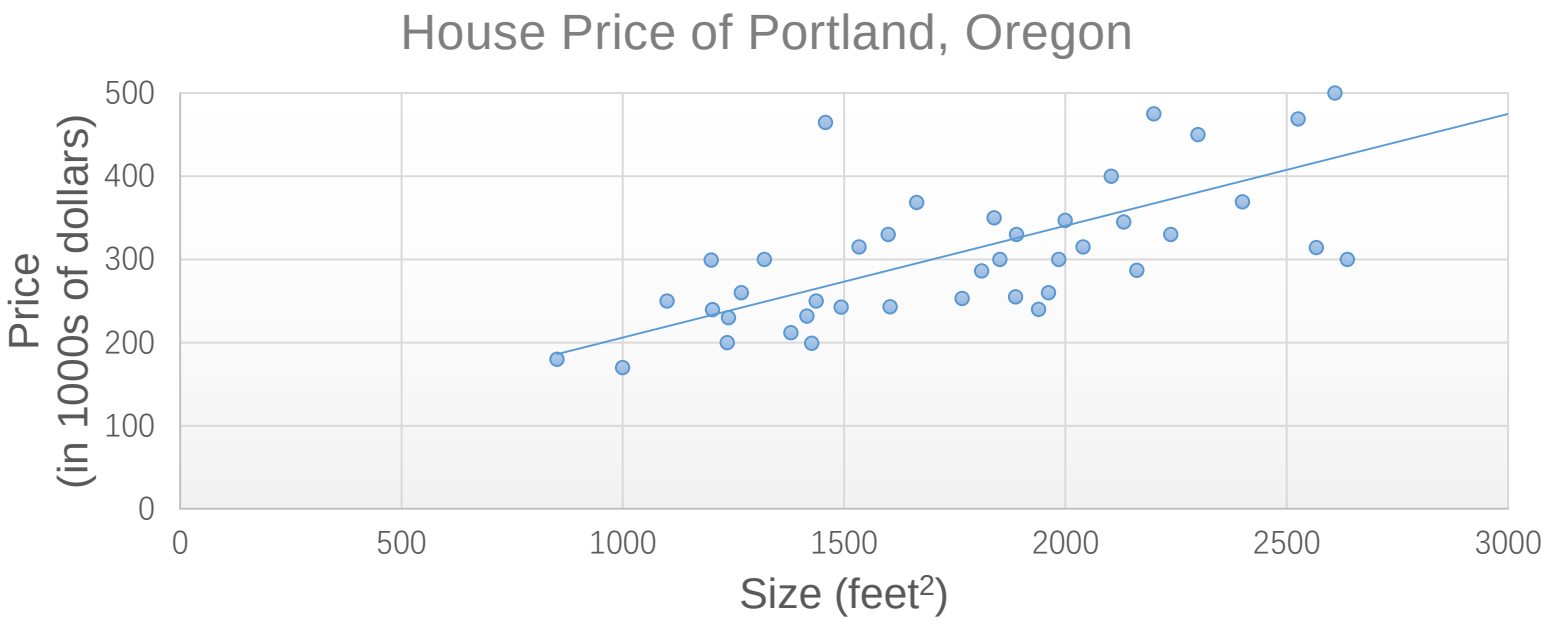
How do we represent f?

Linear Regression: $f(x) = \theta_0 + \theta_1 x$

- The model is in linear in terms of parameters θ_0 (***intercept***) and θ_1 (***coefficient***)
- Linear regression with one variable (univariate linear regression).

We will predict that y is a linear function of x (straight line)

Regression Example



We can predict that the house price is a linear function of house size

Regression Model Evaluation

Given dataset $D = \{(x^{(i)}, y^{(i)}); i = 1,2, \dots, m\}$ and a regression model f, evaluate the performance of the model using following metrics.

Error Metric	Formula	Notes
Mean absolute error (MAE)	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i - f(x_i)\rvert$	Average of the absolute difference between the actual and predicted values.
Mean squared error (MSE)	$\frac{1}{n}\sum_{i=1}^{n}(y_i - f(x_i))^2$	Average of the squared difference between the actual and predicted values.
Root mean squared error (RMSE)	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - f(x_i))^2}$	Square root of Mean Squared error.
R-squared	$1 - \frac{\sum_{i=1}^{n}(y_i - f(x_i))^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	Proportion of the variance for a dependent variable that's explained by the regression model. ***Normally ranges from 0 to 1, the closer to 1 the better performance***

Linear Regression Model Training

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

Model: $f_\theta(x) = \theta_0 + \theta_1 x$

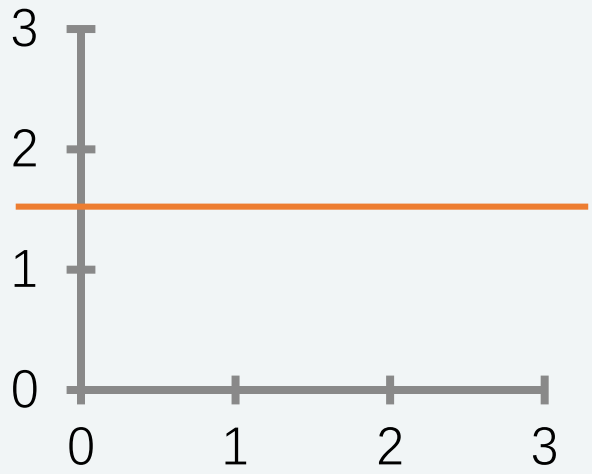
- $\theta_i's$ are the parameters (weights)

Different setting of parameters result in different models, how to choose $\theta_i's$?

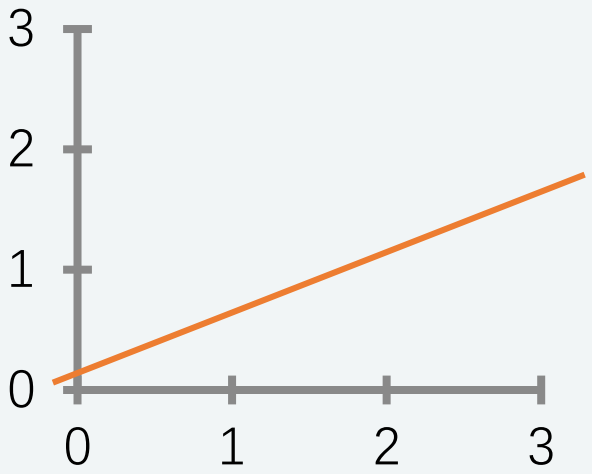
Linear Regression Model Training

Different settings of θ_0 (***intercept***) and θ_1 (***coefficient***) and the corresponding models for

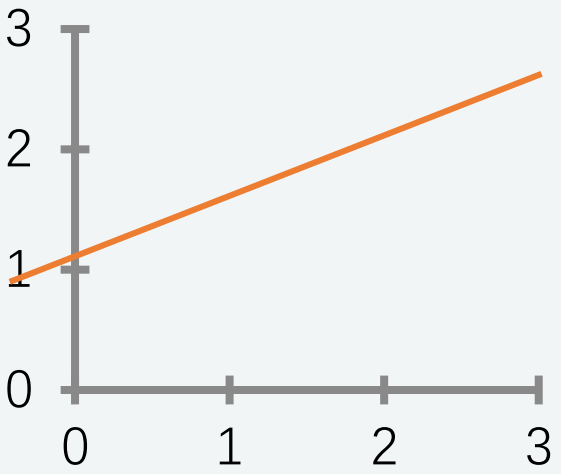
$$f_\theta(x) = \theta_0 + \theta_1 x$$



$\theta_0 = 1.5$
$\theta_1 = 0$



$\theta_0 = 0$
$\theta_1 = 0.5$



$\theta_0 = 1$
$\theta_1 = 0.5$

Multivariate Linear Regression



House Price example with multiple features (variables)

Size in feet^2	Number of bedrooms	Number of floors	Age of home (years)	Price ($) in 1000's
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

Notation

- n: number of features; m: number of training examples
- $x^{(i)}$: input of ith training example
- $x_j^{(i)}$: value of feature j in i^{th} training example
- $y^{(i)}$: output/target of ith training example

Multivariate Linear Regression

House Price example with multiple features (variables)

Size in feet^2	Number of bedrooms	Number of floors	Age of home (years)	Price ($) in 1000's
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

- Univariate Linear Regression (previous): $f_\theta(x) = \theta_0 + \theta_1 x_1$
- Multivariate Linear Regression: $f_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$

To simplify our notation, we also introduce the convention of letting $x_0 = 1$ (this is the intercept term)

- $f_\theta(x) = \sum_{i=0}^{n} \theta_i x_i = \theta^T x$
- on the right-hand, we are viewing θ and x both as vectors

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Model Evaluation and Selection



Generally, the difference between the actual predicted output of the learner and the true output of the sample is called "error"

- Training error/ empirical error: the error of the learner/model on the training data
- Generalization error: the error on the new data

Model Evaluation and Selection

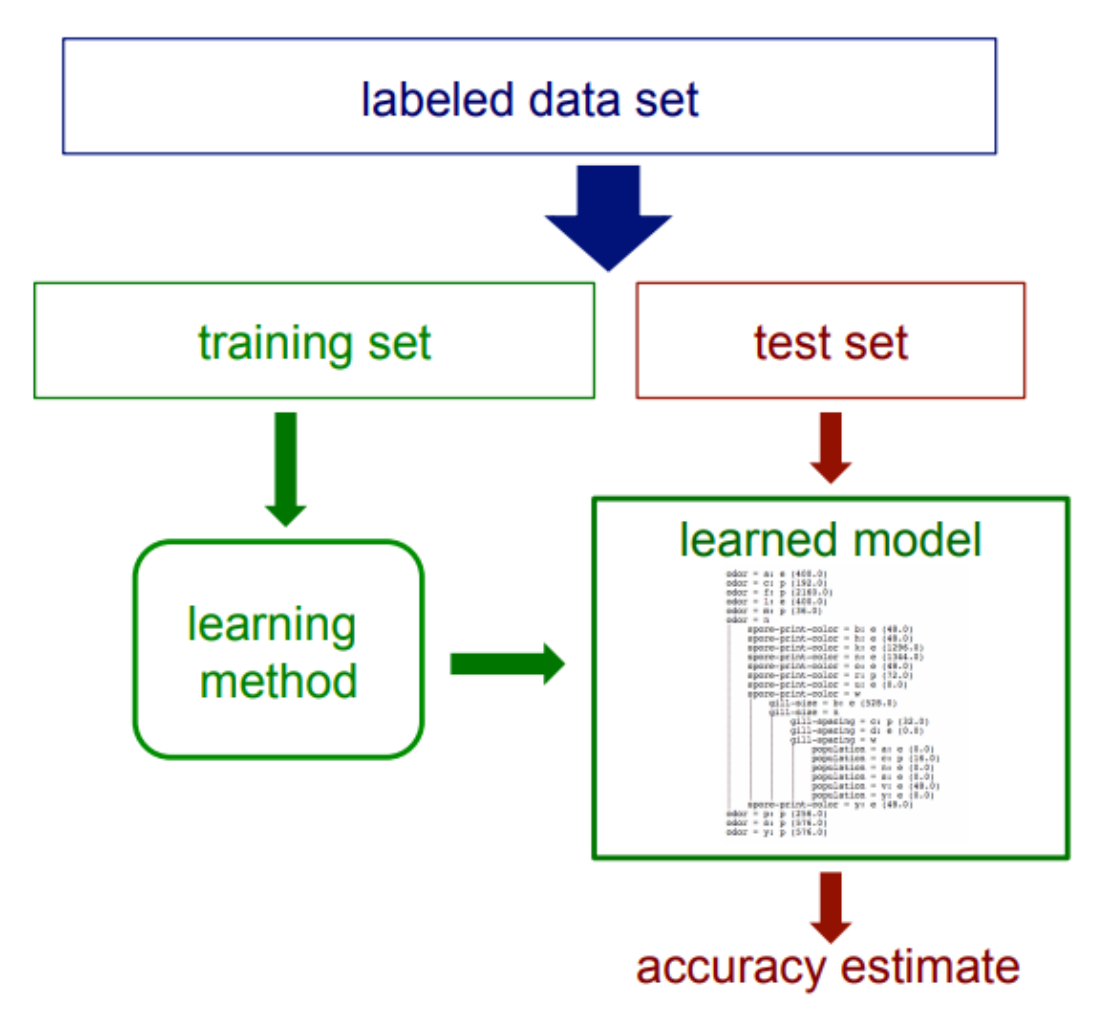


We want to get a learner with a small generalization error

However, we do not have the information for new data, instead we try to minimize the empirical error on the training data

- split data randomly into a training set and a test set (e.g., a 70%/30% split).
- train your model on the training set and see how it performs on the test set.
- use the "testing error" on the test set as an approximation of the generalization error

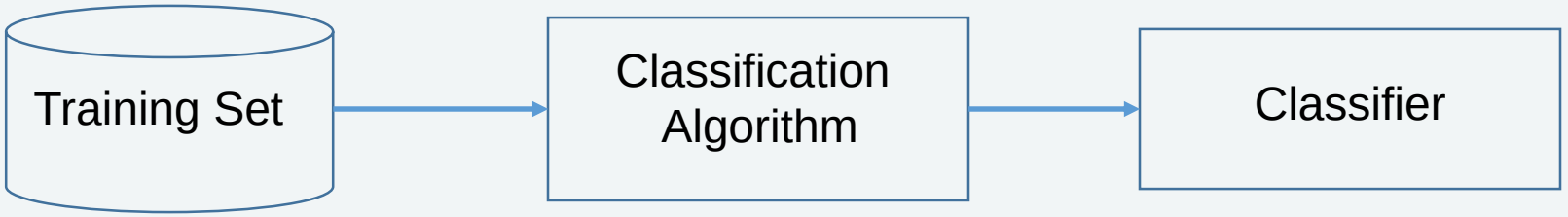
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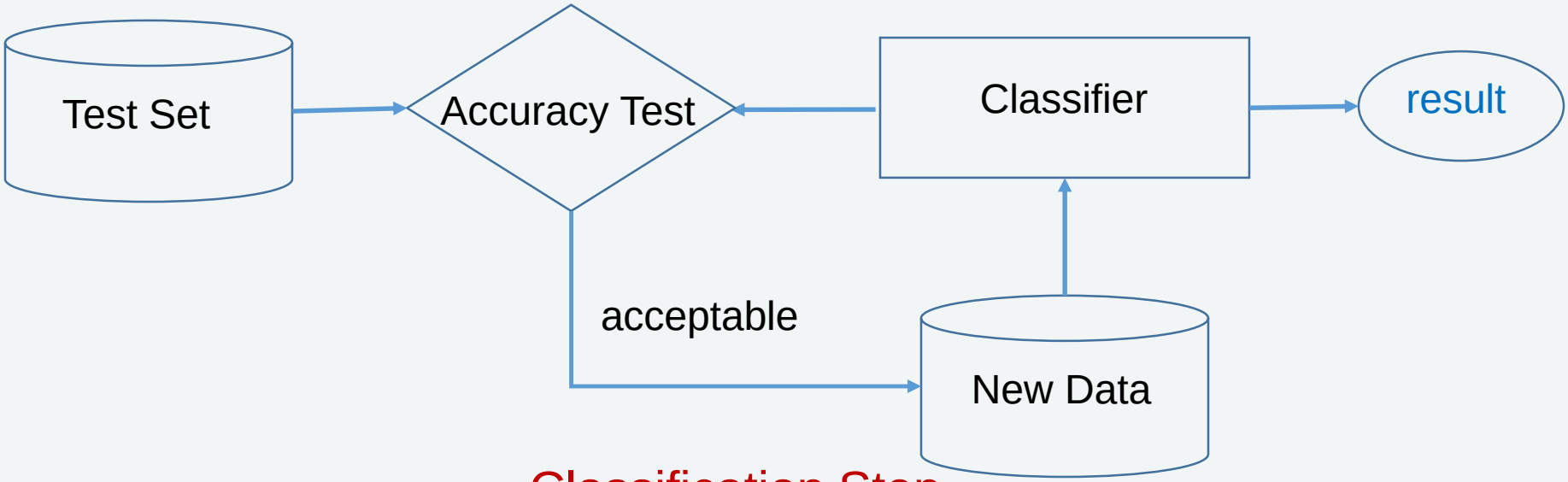
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- Clustering Method (Unsupervised Learning)
  - Objective
  - Similarity Measures
  - Method 1: Hierarchical Clustering
  - Method 2: K-Means Method (Clustering by Partitioning)
- Lab (Demo): Unsupervised Learning

How to Perform Classification?



Learning Step



Classification Step

Classification Accuracy



How to measure the accuracy of a classifier?

- Suppose we have already selected the training and test sets, and we have created a classifier based on the training set

We will measure the accuracy based on the test set

- Try to predict the class value of every tuple in the test set, and compare it against their actual ones (which are already stored in the test set)

Accuracy Measures

Classification accuracy

- The percentage of test set tuples that are correctly classified by the classifier
- A useful tool for analyzing how well the classifier can recognize tuples of different classes is the confusion matrix

c_{ij}: number of tuples from class *i* that are classied as class *j* by the classifier

Predicted class

accuracy of classifying "yes" tuples

Actual class

Classes	Buy="yes"	Buy="no"	Total	Accuracy
Buy="yes"	6,954	46	7000	99.34
Buy="no"	412	2,588	3000	86.27
Total	7,366	2,634	10,000	95.42

total accuracy

Accuracy Measures

Classification accuracy sometimes can be misleading

Let us focus on a two-class problem (e.g., “non cancer”/ “cancer" patients) where

- number of class C_1 tuples: 9,990
- number of class C_2 tuples: 10

- If the classifier predicts everything to be class C_1, then accuracy is 99.9%
- However, this is misleading because the classifier does not correctly predict any tuple from C_2

Accuracy Measures

Consider a two-class problem and the confusion matrix below

- The positives refers to the tuples of the main class of interest (C_1)

Predicted class

Actual class

Classes	C_1	C_2
C_1	true positives (TP)	false negatives (FN)
C_2	false positives (FP)	true negatives (TN)

Accuracy Measures (COVID test)

Classes	*You tested positive*	*You tested negative*
You have COVID	true positives (TP) 真阳	false negatives (FN) 假阴
You don't have COVID	false positives (FP) 假阳	true negatives (TN) 真阴

Accuracy Measures (Face recognition)

Classes	*Face regognized* ***positive***	*Face recognized* ***negative***
Yourself are in front of device	true positives (TP) 真阳	false negatives (FN) 假阴
Yourself aren't in front of device	false positives (FP) 假阳	true negatives (TN) 真阴

Accuracy Measures

- **Accuracy** is a measure for how many correct predictions your model made for the complete test dataset. It is measured by the following formula:

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Accuracy Measures

- **Recall** – or the true positive rate – is the measure for how many true positives get predicted out of all the positives in the dataset. It is sometimes also called the **sensitivity**. The measure is collected by the following formula:

$$recall = \frac{TP}{TP+FN}$$

Accuracy Measures

- **Precision** is a measure for the correctness of a positive prediction. In other words, it means that if a result is predicted as positive, how sure can you be this is actually positive. It is calculated using the following formula:

$$precision = \frac{TP}{TP+FP}$$

Accuracy Measures

- An f-score is a way to measure a model's accuracy based on recall and precision. There's a general case F-score, called the **F1-score**
- The higher an F-score, the more accurate a model is. The lower an F-score, the less accurate a model is.

$$F1 = 2 \cdot \frac{precision \cdot recall}{precision + recall}$$

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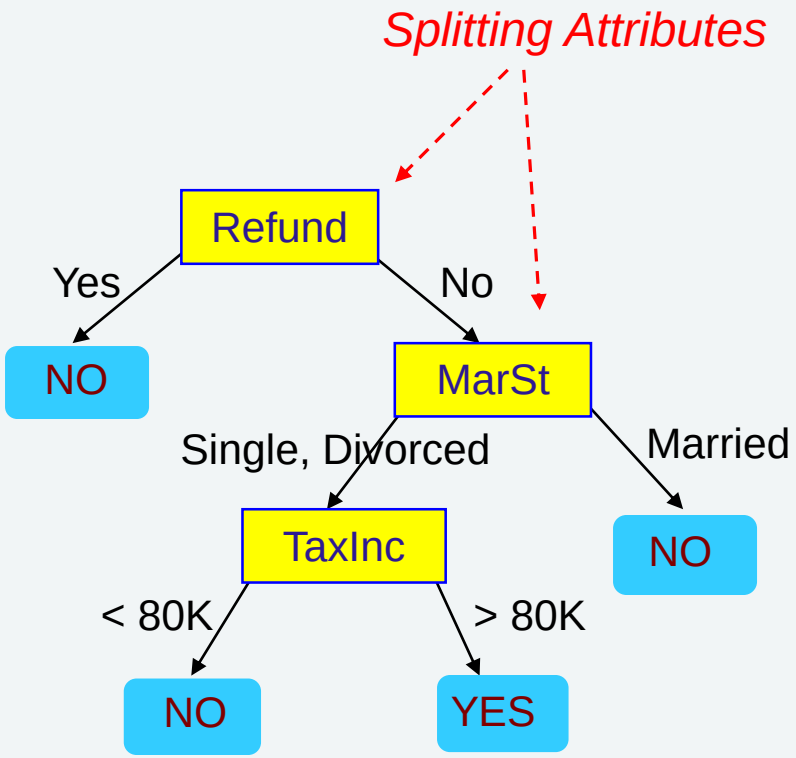
Decision Tree



- One way to solve the classification problem is by asking a series of questions about the attributes of the test record
- Each time we receive an answer, a follow-up question is asked until we reach a conclusion about the class label of the record
- The series of questions and their possible answers can be organized in the form of a decision tree, which is a hierarchical structure consisting of nodes and directed edges

Decision Tree

- Each internal node denotes a test on an attribute
- Each branch represents an outcome of the test
- Each leaf node holds a class label

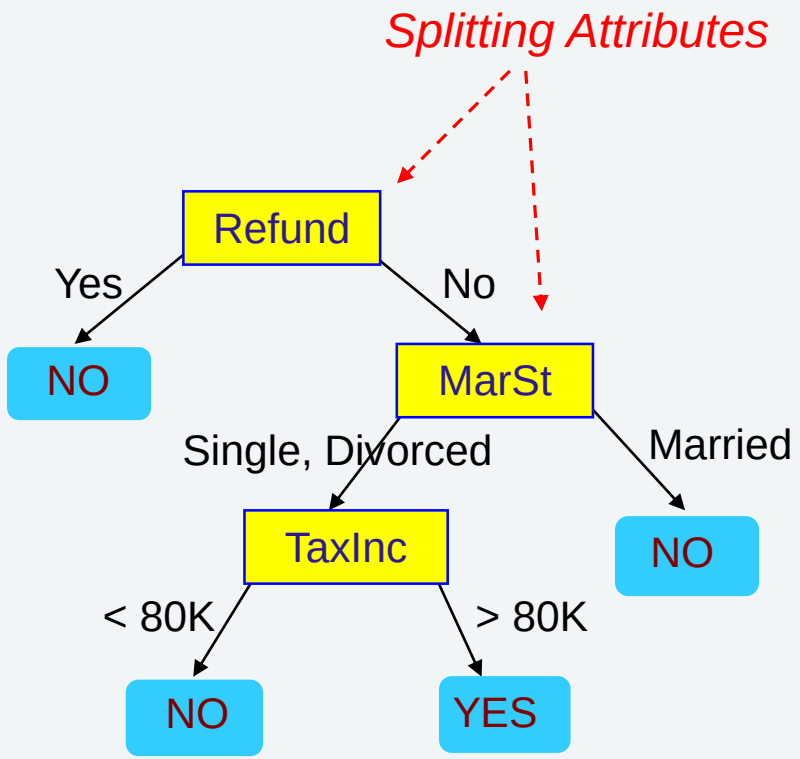


Model: Decision Tree

Decision Tree

There could be more than one tree that fits the same data!

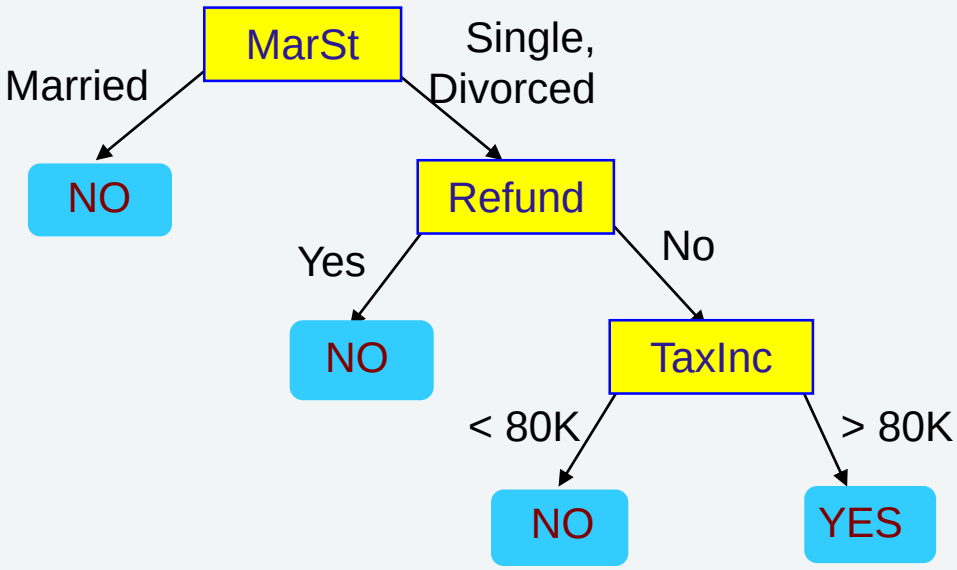
Tid	Refund (categorical)	Marital Status (categorical)	Taxable Income (continuous)	Cheat (class)
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Decision Tree

There could be more than one tree that fits the same data!

Tid	Refund	Marital Status	Taxable Income	Cheat
	categorical	categorical	continuous	class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Decision Tree



COVID-19 DECISION/EXPOSURE TREE

ARE YOU EXPERIENCING COVID-19 SYMPTOMS?

ANY **ONE** OF THESE SYMPTOMS

> Fever equal to or over 100.4 F
(Without any fever-reducing medications, acetaminophen, or ibuprofen.)

> New onset dry cough

> Loss of smell or taste

> Shortness of Breath / Difficulty Breathing

ANY **COMBINATION** OF THESE SYMPTOMS WITH OR WITHOUT SYMPTOMS TO THE LEFT

> Chills

> New or unusual headache

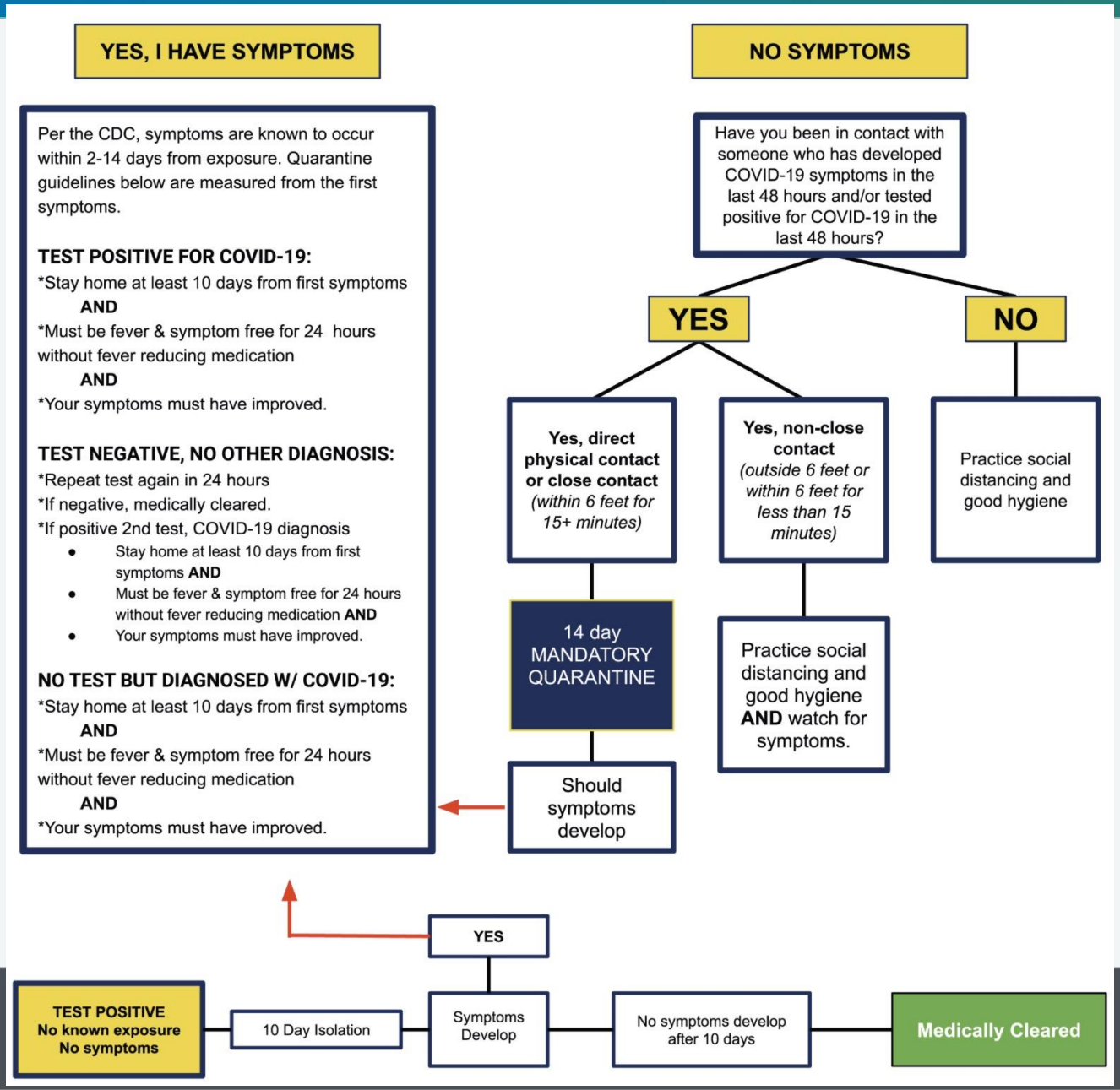
> Nausea, vomiting, diarrhea, or loss of appetite

> Muscle aches

> Sore throat

> Congestion/runny nose

Decision Tree



Decision Tree: Prediction

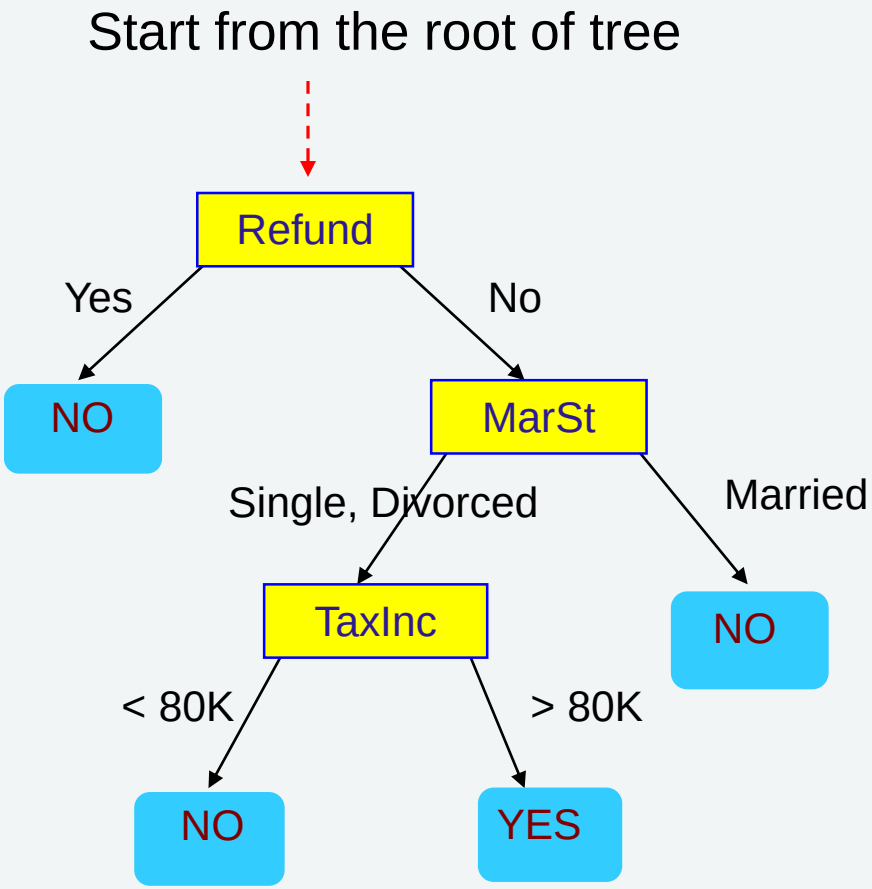
Given a new (unseen) tuple: the associated class is unknown

- Test the attribute values of the tuple against the decision tree

  Start from the root and trace a path to a leaf node (top-down), based on the attribute values of the tuple

- The class value included in this leaf is assigned to the tuple

Decision Tree: Prediction



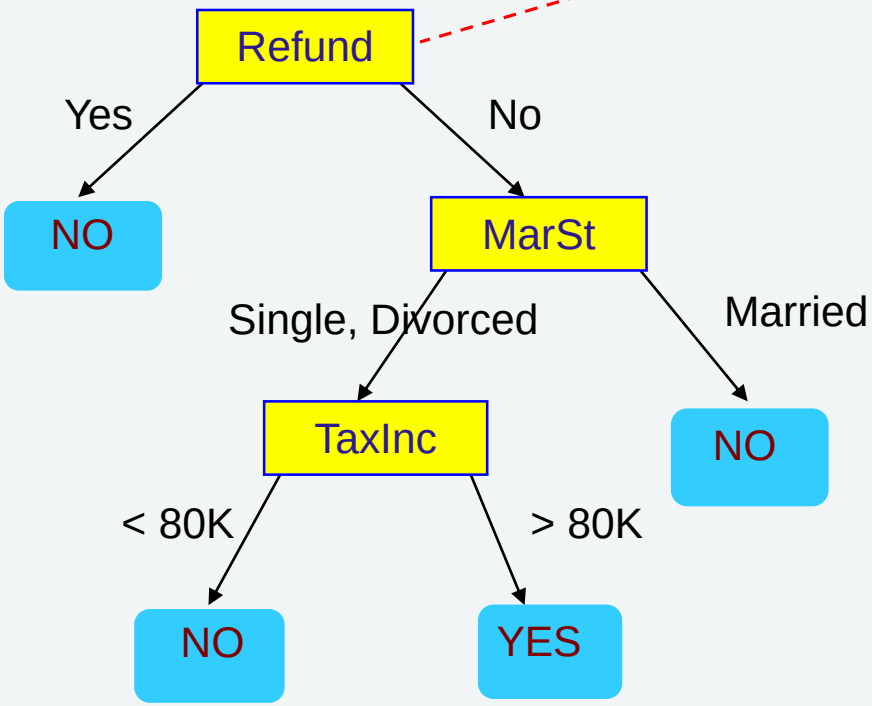
Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

Decision Tree: Prediction

Test Data

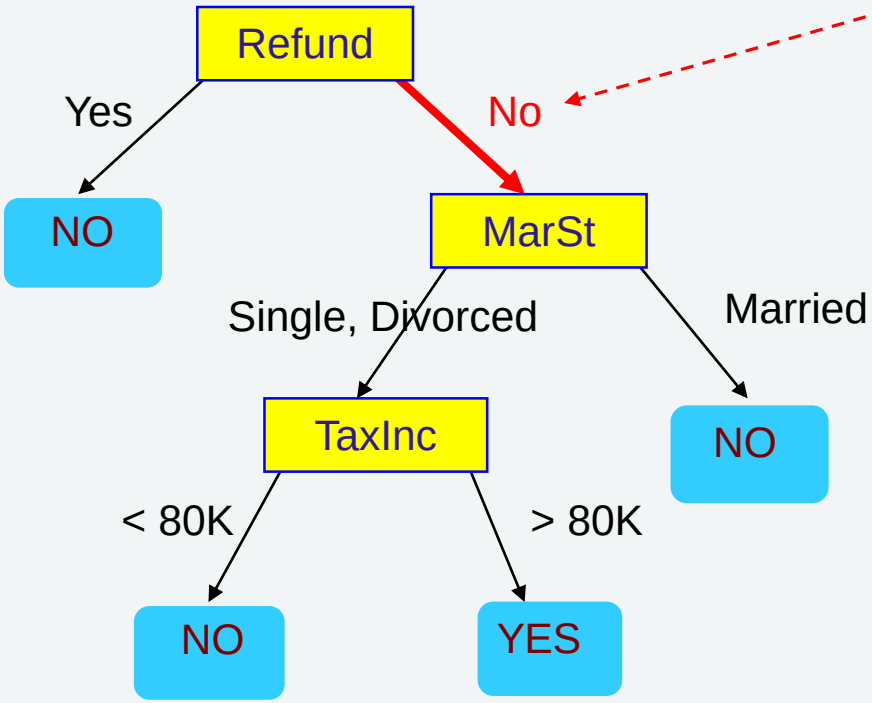
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Decision Tree: Prediction

Test Data

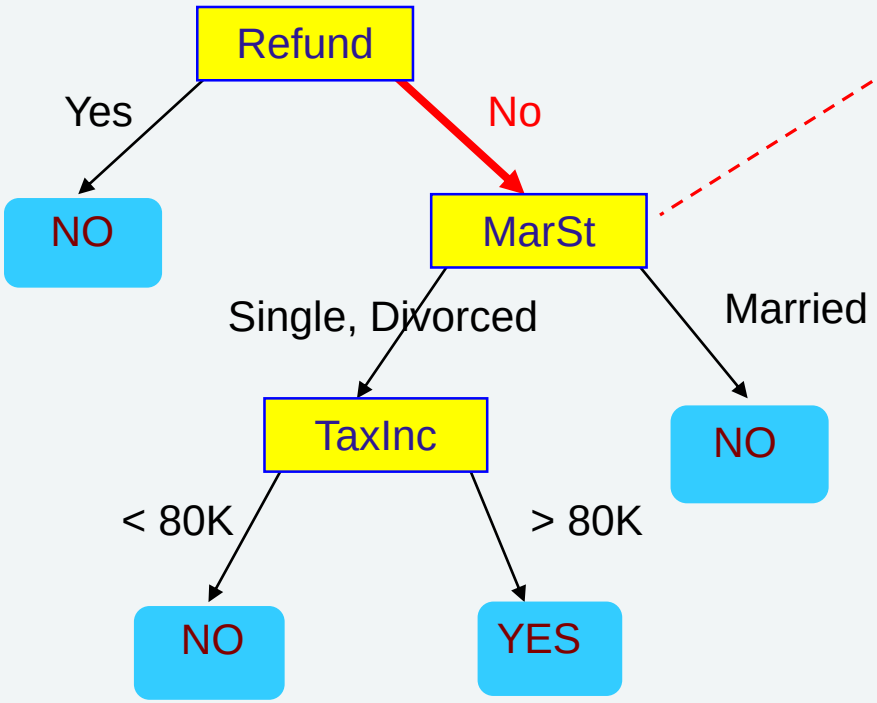
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Decision Tree: Prediction

Test Data

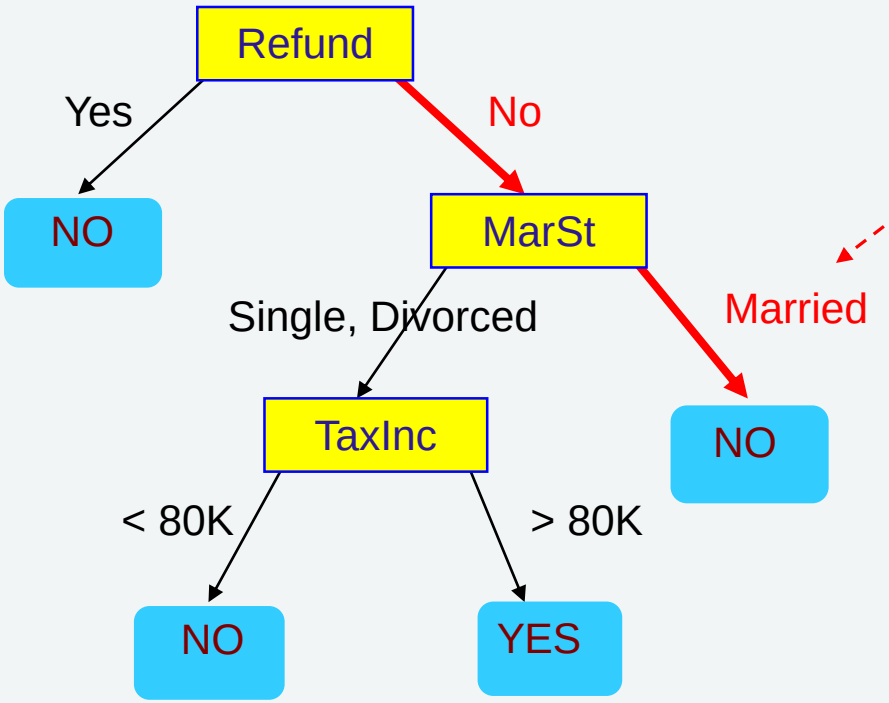
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



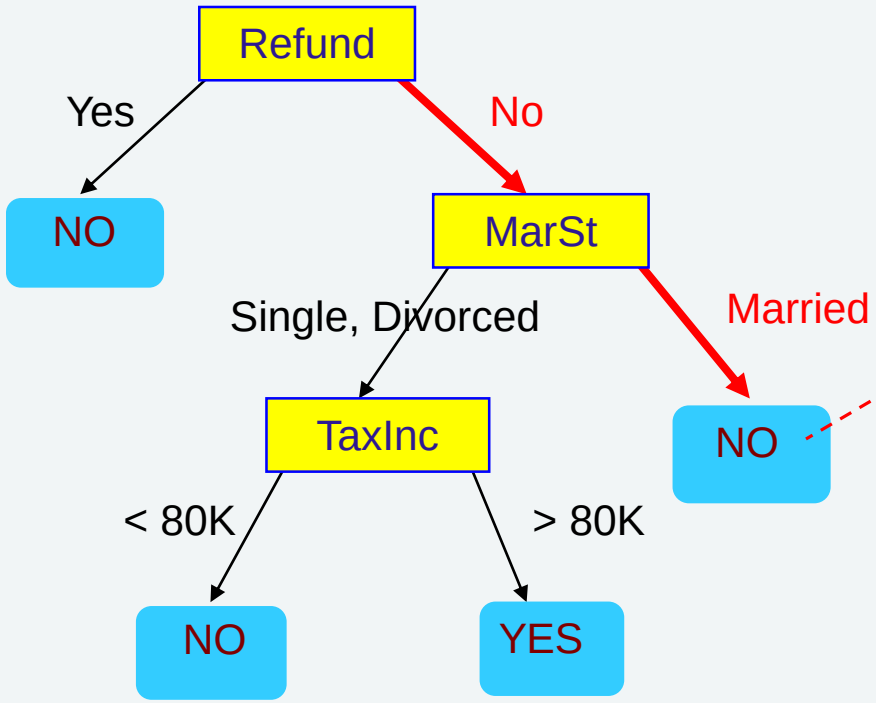
Decision Tree: Prediction

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Decision Tree: Prediction



Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

Assign Cheat to “No”

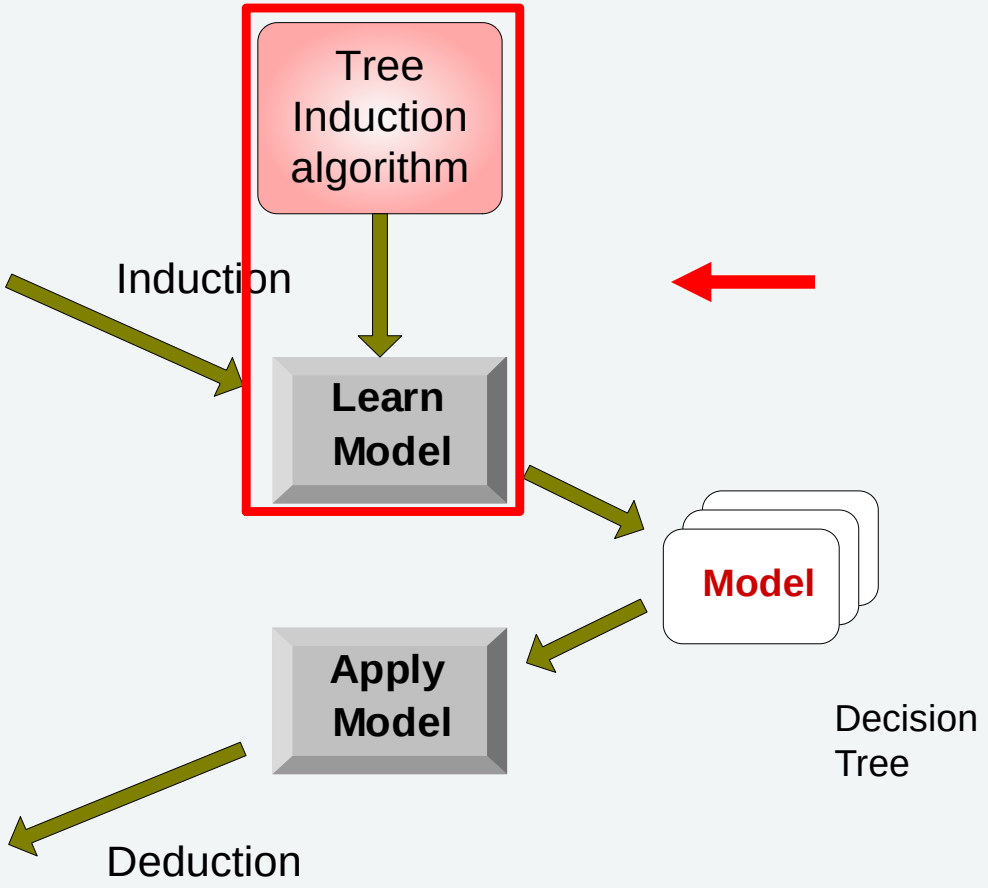
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



How to Build A Decision Tree



- There are exponentially many decision trees that can be constructed from a given set of attributes. While some of the tree are more accurate than others
- Efficient algorithms have been developed to induce a reasonably accurate decision tree in a reasonable amount of time
- These algorithms usually employ a greedy strategy that grow a decision tree by making a series of locally optimum decisions about which attribute to use for partitioning the data

Decision Tree Induction



- Basic algorithm (a greedy algorithm)
  - Tree is constructed in a top-down recursive divide-and-conquer manner
- Input
  - Node N, the first time the algorithm is called, root of the tree
  - Dataset D of training examples
    Initially, entire training set
  - Attribute list, holds a set of attributes
    Initially the attributes that remained after data preprocessing
  - Attribute selection method: a heuristic process for selecting the attribute that "best" discriminates the given tuples according to class
- Output
- Decision tree

If you are CDS, you will learn in Y2/Y3

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Goal of Cluster Analysis

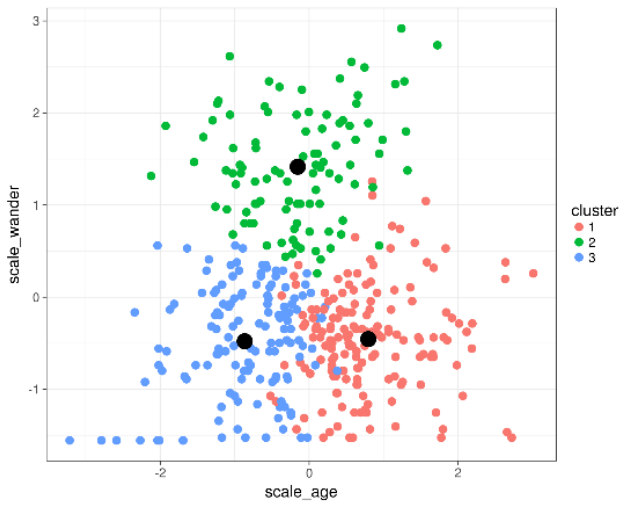
Discover hidden structures in unlabeled data (unsupervised)

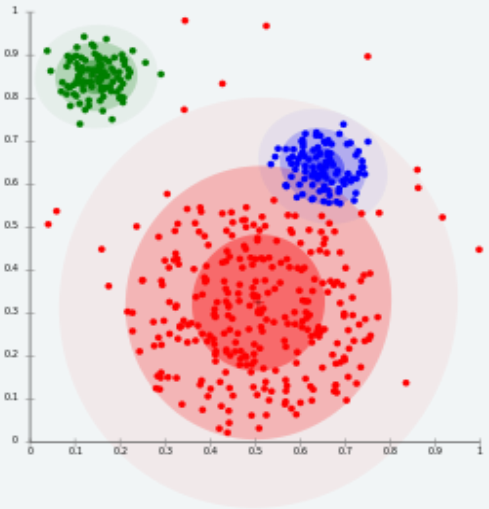
Clustering identifies a finite set of groups (*clusters*) $C_1, C_2 \cdots, C_k$ in the dataset such that:

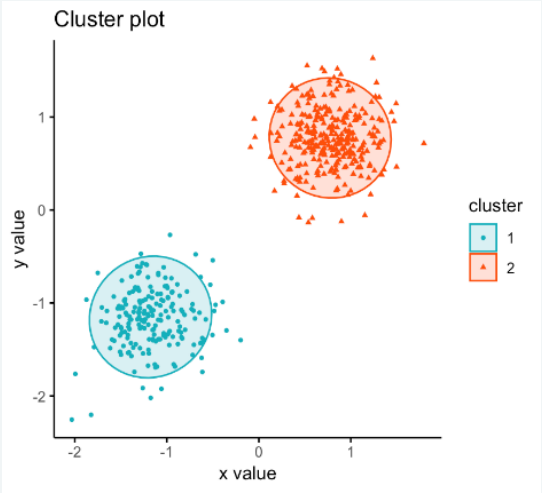
- Objects within the *same* cluster C_i shall be as similar as possible
- Objects of *different* clusters C_i, C_j $(i \neq j)$ shall be as dissimilar as possible

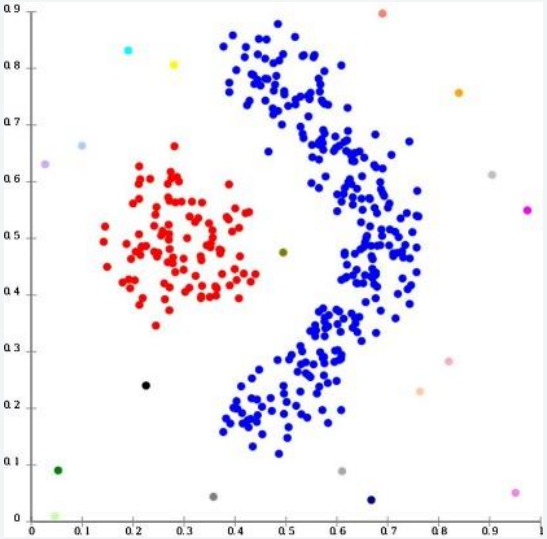
Cluster Properties

- Clusters may have different sizes, shapes, densities
- Clusters may form a hierarchy
- Clusters may be overlapping or disjoint









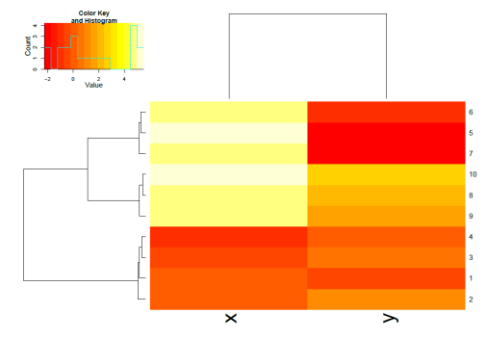
Application Examples



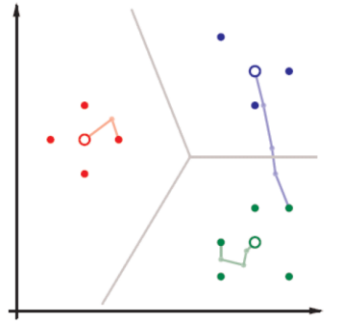
- **Customer segmentation**
  - Find groups of customers with similar behaviour; find customers with unusual behavior
- **Molecule search**
  - Find molecules with similar structure to already working ones
- **Anomaly detection**
  - Find unusual patterns in data from sensors monitoring mechanical engines
- **Determining user groups on the WWW**
  - *Clustering of activities in web-logs*
  - *Find groups of social media users with similar attitude.*
- **Structuring large sets of text documents**
  - *hierarchical clustering of the text documents*
- **Generating thematic maps from satellite images**
  - *clustering sets of raster images of the same area (feature vectors)*

Types of Clustering Approach

Linkage Based
e.g. Hierarchical Clustering



Clustering by Partitioning
e.g. k-Means



We will use those two approaches only

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(Dis-)similarity Functions for Numeric Attributes

For two objects $\boldsymbol{x} = (x_1, x_2, \cdots, x_d)$ and $\boldsymbol{y} = (y_1, y_2, \cdots, y_d)$:

- ***Minkowski-Distance (L_p-Metric)***

$$d_p(\boldsymbol{x}, \boldsymbol{y}) = \sqrt[p]{\sum_{i=1}^{d} |x_i - y_i|^p}$$

- ***Euclidean Distance ($L_2 - p = 2$)***

$$d_E(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$$

- ***Manhattan-Distance ($L_1 - p = 1$)***

$$d_M(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{d} |x_i - y_i|$$

- *Tschebyschew-Distance ($L_\infty - p = \infty$)*

$$d_\infty(\boldsymbol{x}, \boldsymbol{y}) = \max_{1 \le i \le d} \{|x_i - y_i|\}$$

- *Cosine Distance*

$$d_C(\boldsymbol{x}, \boldsymbol{y}) = 1 - \frac{\boldsymbol{x}^T \boldsymbol{y}}{\|\boldsymbol{x}\| \|\boldsymbol{y}\|}$$

- *Tanimoto Distance*

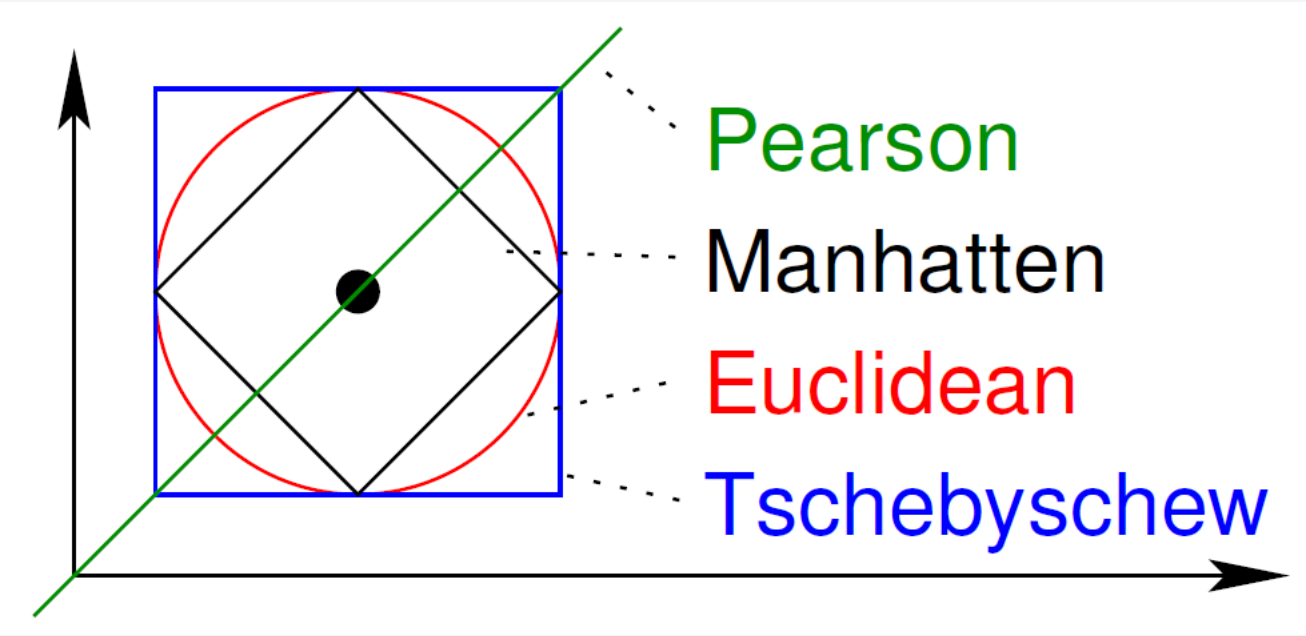
$$d_T(\boldsymbol{x}, \boldsymbol{y}) = 1 - \frac{\boldsymbol{x}^T \boldsymbol{y}}{\|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 - \boldsymbol{x}^T \boldsymbol{y}}$$

- *Pearson Distance*

Euclidean distance of z-score transformed $\boldsymbol{x}, \boldsymbol{y}$

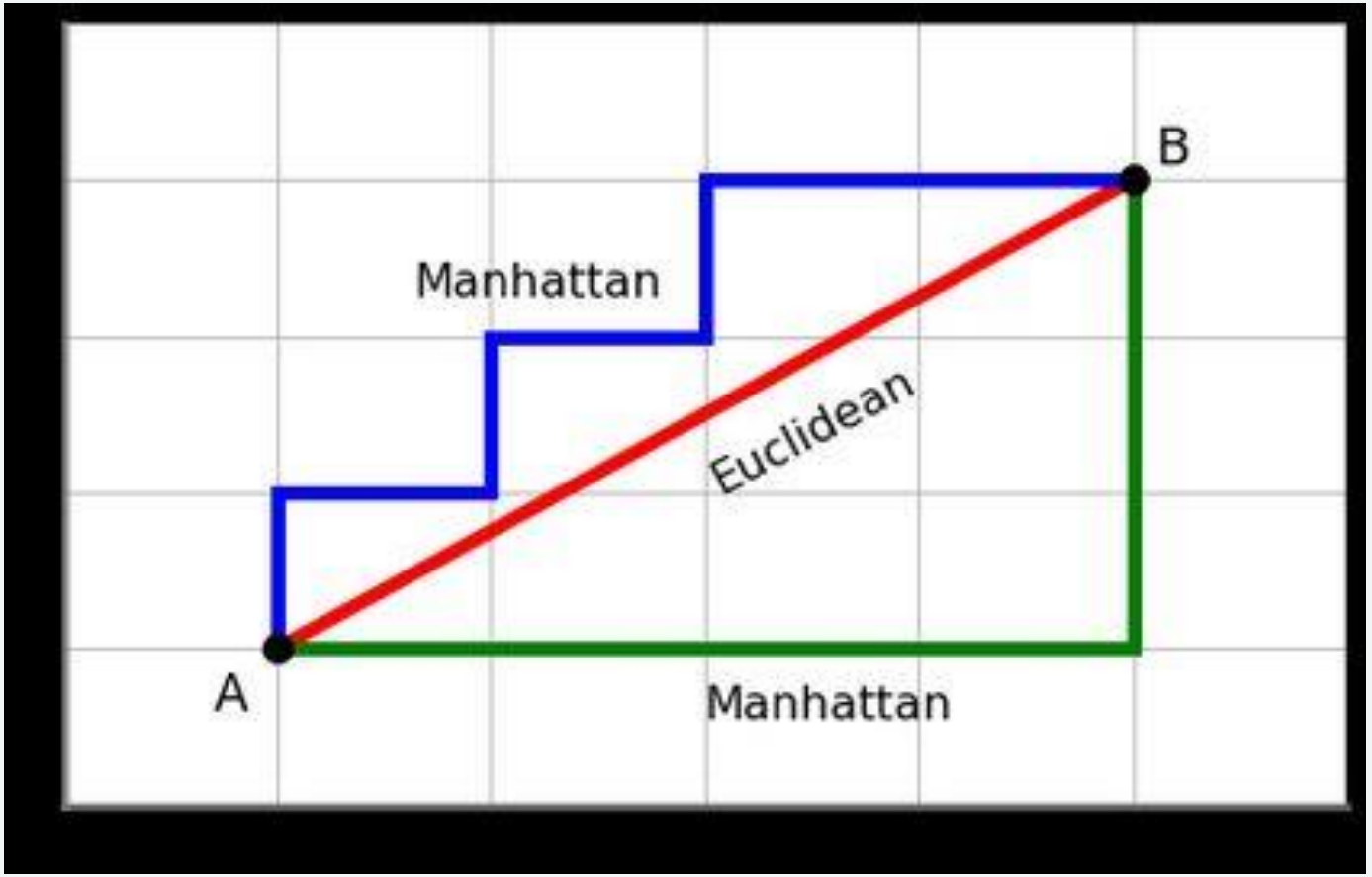
(Dis-)similarity Functions for Numeric Attributes

Various choice of dis-similarity between two numerical vectors



(Dis-)similarity Functions for Numeric Attributes

Various choice of dis-similarity between two numerical vectors



Influence of Distance Function / Similarity

- Clustering vehicles:
  - red Ferrari
  - green Porsche
  - red Bobby car





- Distance function based on maximum speed (numeric distance function):
  - Cluster 1: Ferrari & Porsche
  - Cluster 2: Bobby car

The distance function affects the shape of the clusters

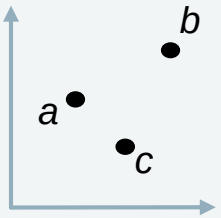
- Distance function based on color (nominal attributes):
  - Cluster 1: Ferrari and Bobby car
  - Cluster 2: Porsche

Similarities, Dissimilarities, and Distances

Given data points, how can we summarize how different they are? (Rather than summarizing the similarity)

- Dissimilarity metric – ***distance***
- Distance matrix – pairwise differences of all data points
- Distance $d_{i,j}$ (between i and j) calculated as the Euclidean distance

id	X	Y
a	1	2
b	3	3
c	2	1



$[d_{i,j}]$	a	b	c
a	0.00	2.23	1.41
b	2.23	0.00	2.23
c	1.41	2.23	0.00

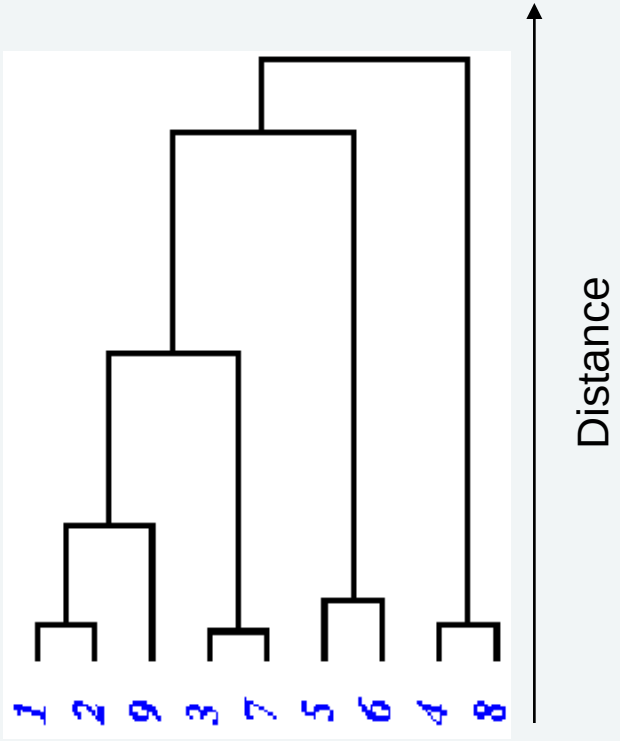
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Linkage Hierarchies: Basics

Goal

- Construction of a hierarchy of clusters (*dendrogram*) by merging/separating clusters with minimum/maximum distance

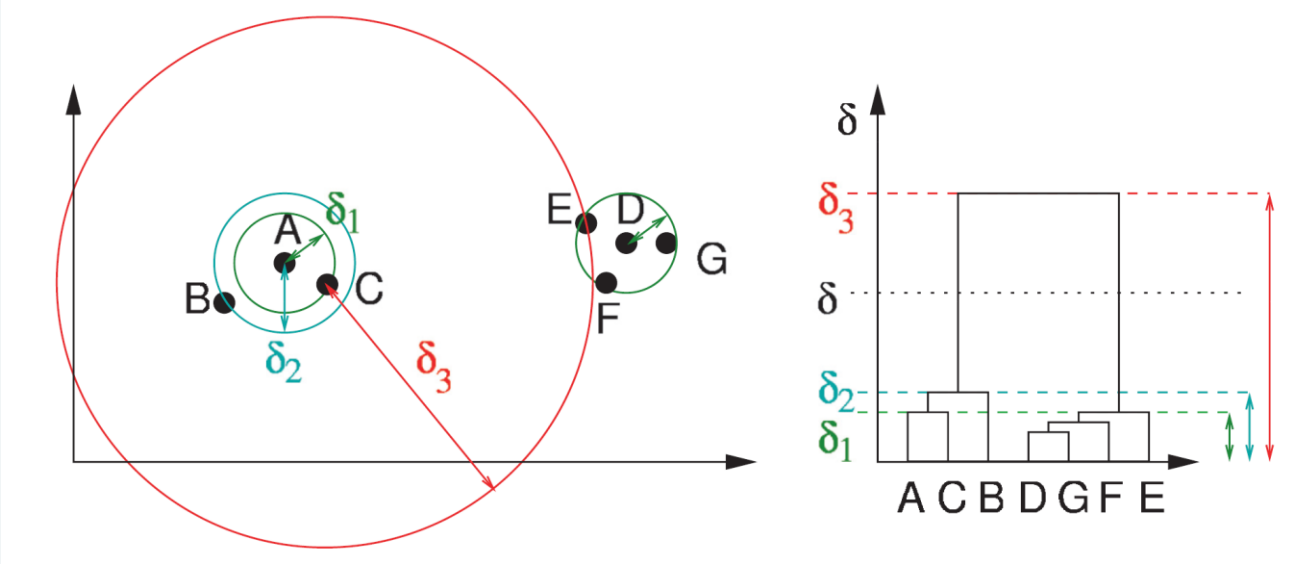


Dendrogram:

- A tree representing a hierarchy of clusters, with the following properties:
  - Root: single cluster with the whole data set.
  - Leaves: clusters containing a single object.
  - Branches: merges / separations between larger clusters and smaller clusters / objects

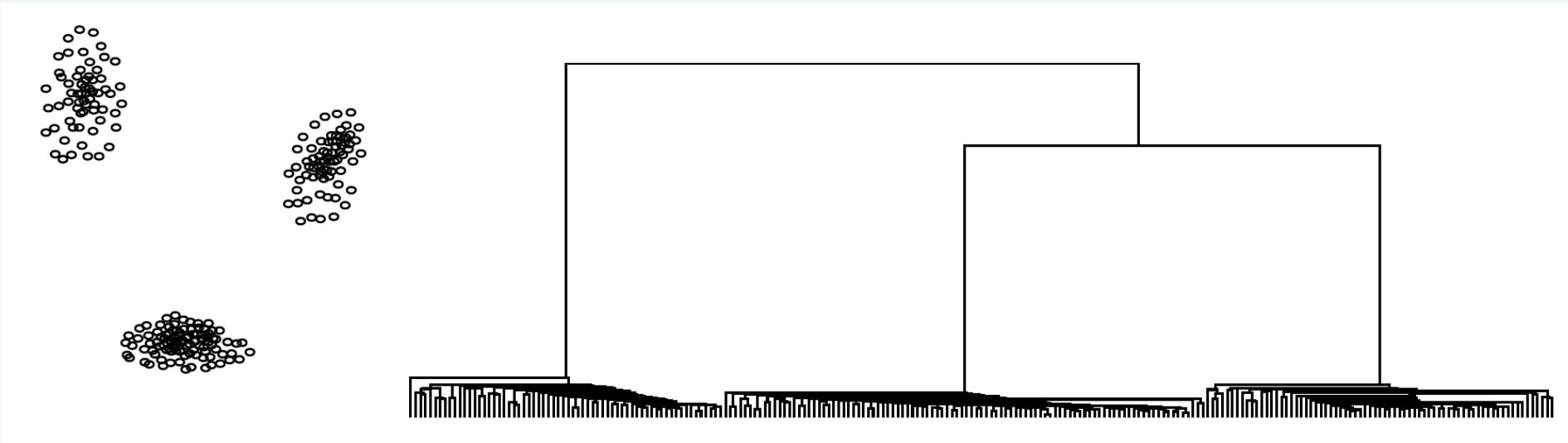
Hierarchy of Clusters

The hierarchy of the clustering is described by a ***dendrogram***



- At $\delta = 0$ ➔ 7 clusters of singletons
- At $\delta = \mathrm{d(D, G)}$ ➔ D & G form a cluster, all others are singleton clusters
- At $\delta = \mathrm{d(A, C)}$ ➔ A & C form a cluster, B remains a singleton, D, E, F, & G is another cluster
- And so on

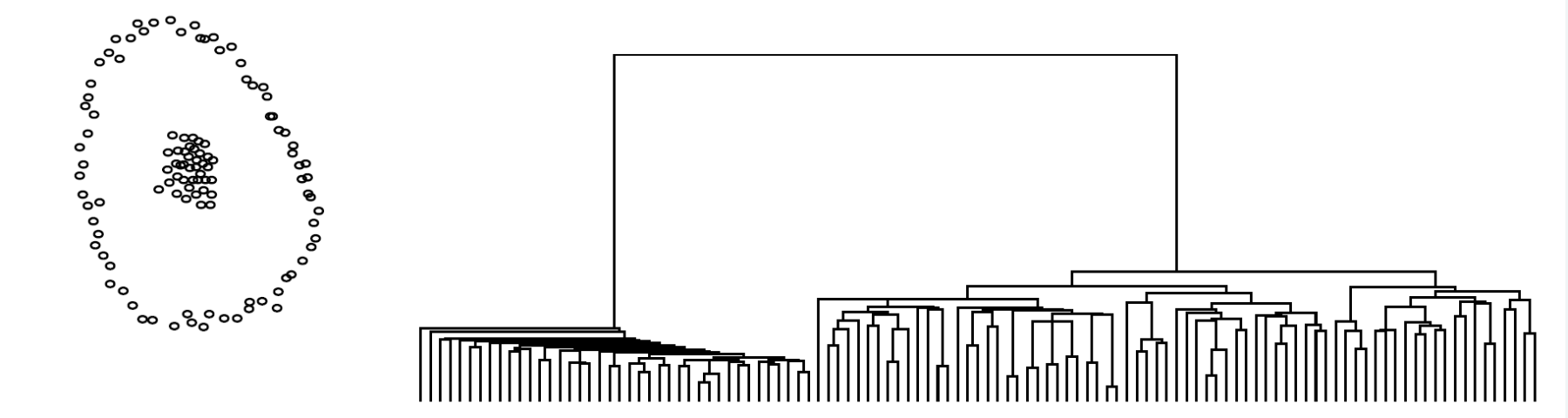
Hierarchy of Clusters (Examples)



In this example:

- Three well-separated clusters
- Formed with small δ
- Remains stable for a wide range of δ (until large δ)

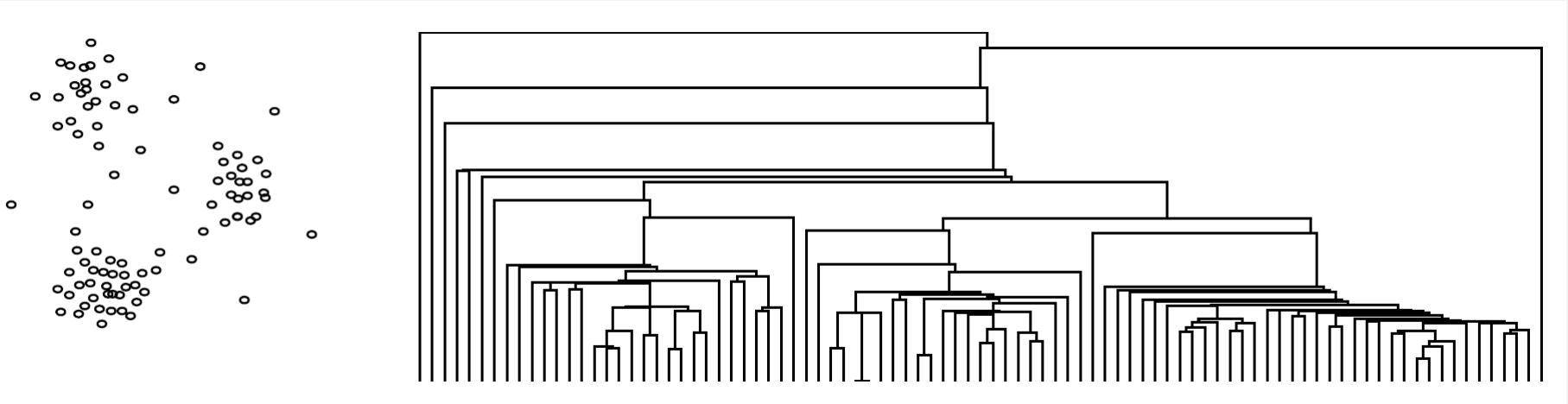
Hierarchy of Clusters (Examples)



In this example:

- Two well-separated clusters
- Clusters differ in sizes and shapes

Hierarchy of Clusters (Examples)



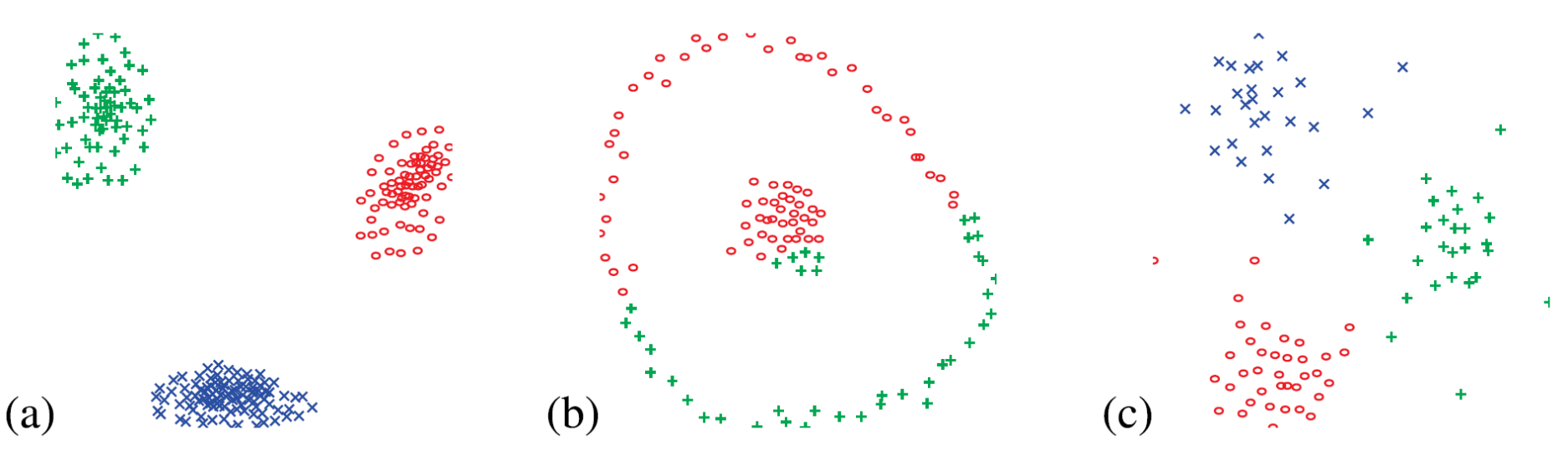
In this example:

- Small noises to clusters
- Isolated noise points are ***chained*** – forming clusters with small δ
- No clearly defined robust clusters
- Hierarchical clustering (esp. single-linkage) is susceptible to noises

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- **Clustering Method (Unsupervised Learning)**
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  - Similarity Measures
  - Method 1: Hierarchical Clustering
  - **Method 2: K-Means Method (Clustering by Partitioning)**
- Lab (Demo): Unsupervised Learning

Overview of Partitioning



- Works well when the assumption is true (i.e. closest prototype represents a cluster)
- But does not work all cases – e.g., circular clusters

Partitioning

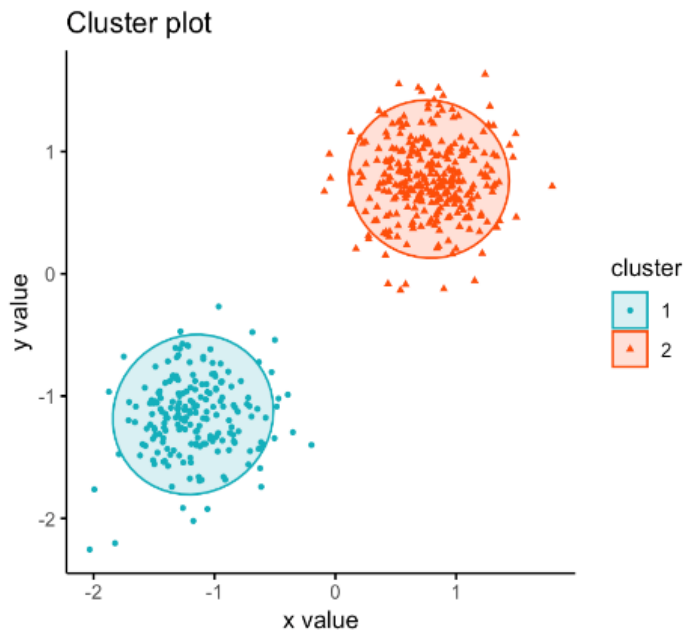
Goal: A (disjoint) partitioning into k clusters with minimal costs

Local optimization method:

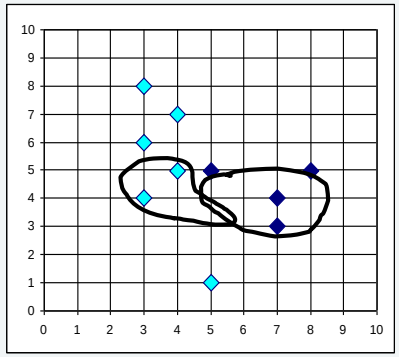
- choose k initial cluster representatives
- optimize these representatives iteratively
- assign each object to its **most similar cluster representative**

Types of cluster representatives:

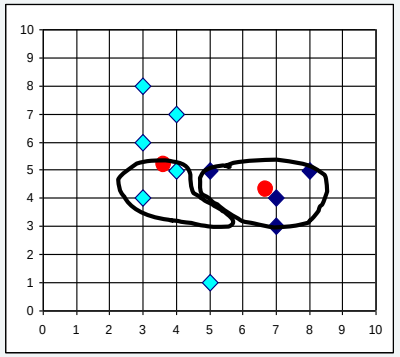
- Mean of a cluster (*construction of central points*)
- Median of a cluster (*selection of representative points*)
- Probability density function of a cluster (*expectation maximization*)



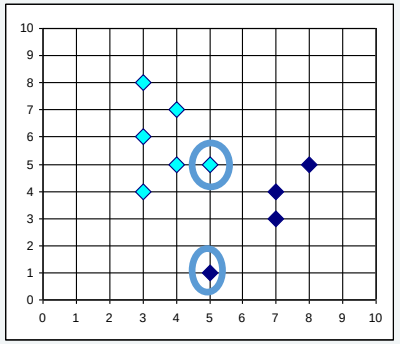
k-Means Algorithm



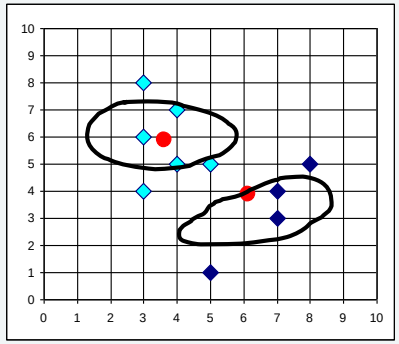
Calculation of new centroids



Cluster assignment



Calculation of new centroids



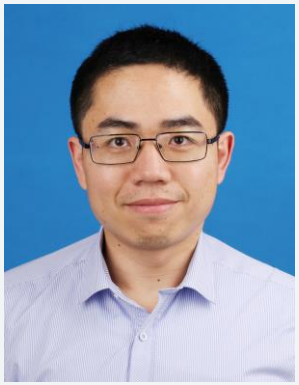
If you are CDS, you will learn in Y3

Clustering and face recognition



- Deep learning: face recognition
- All the faces are mapped into non-overlapping cluster of radius **0.6 (means distance < 0.6 is a match)**

Same person?



```
[-0.10896184  0.05544664  0.06675301 -0.0031338  -0.04224422 -0.06632324
 -0.09370329 -0.10551011  0.08113094 -0.06553721  0.27272737 -0.01903389
 -0.21348667 -0.15376335 -0.03805079  0.20436208 -0.19817452 -0.12764442
 -0.02354575  0.01954814  0.07374688 -0.04928869  0.07703349  0.03434119
 -0.15626766 -0.32539505 -0.09599499 -0.1018362  -0.03666351 -0.03615996
 -0.08314003  0.02546445 -0.17006941 -0.04386618  0.03059519  0.07935175
  0.02606429 -0.06789137  0.20701931  0.01349168 -0.19135325  0.0149933
  0.05353317  0.25599992  0.20805822  0.09332568  0.04845903 -0.10678323
  0.15977696 -0.22658752  0.05239259  0.14048949  0.05020547  0.05772729
 -0.01779214 -0.1135909   0.06799653  0.04007015 -0.09515589 -0.02561913
  0.09202795 -0.00407893 -0.00891332 -0.06910229  0.22586182  0.06615839
 -0.13247573 -0.10960731  0.06510356 -0.10948805 -0.07080921  0.05313661
 -0.21889403 -0.16474484 -0.34336478 -0.00633985  0.40432549  0.08666191
 -0.1697778  -0.01728107 -0.07597457  0.02323727  0.13430965  0.08446945
 -0.01344115  0.00402731 -0.1439795   0.01987924  0.27384663 -0.09059483
 -0.02993309  0.18410836  0.02972636  0.09610896  0.00531289  0.0674793
 -0.02276792  0.04800851 -0.12424111  0.04562626  0.05975612  0.02071599
  0.00834936  0.09229372 -0.14192261  0.06103314  0.02019319  0.04037908
  0.02578369  0.00994095 -0.17506778 -0.08651195  0.07921262 -0.16465698
  0.17953324  0.20761904  0.07307686  0.04870486  0.11930457  0.08561553
 -0.08019133  0.0082659  -0.15669098 -0.04434975  0.08815156  0.02454746
  0.10166635 -0.01058803]
```



```
array([-0.09133875,  0.03444508,  0.02782696,  0.00994523, -0.19686258,
        0.02486066,  0.0335204 , -0.03233519,  0.05744473,  0.00501723,
        0.27416849, -0.04127728, -0.2849327 , -0.02508496, -0.05668204,
        0.09556217, -0.1481169 , -0.09453708, -0.1419163 , -0.13951337,
        0.04741215,  0.01541864, -0.05834462,  0.02059731, -0.18354006,
       -0.22195035,  0.02266514, -0.02640289,  0.11226526, -0.17144683,
        0.06336956,  0.16527608, -0.18186924, -0.08615917,  0.06421676,
        0.13525604, -0.05836667, -0.0215878 ,  0.23250441, -0.02983644,
       -0.14385158,  0.12547699,  0.10961694,  0.31671011,  0.1699792 ,
        0.0203779 , -0.00561961,  0.00048708,  0.09413071, -0.32722723,
       -0.02708439,  0.18948245,  0.17577519,  0.07000284, -0.00277642,
       -0.16941926, -0.02865946,  0.13519521, -0.16064328,  0.13494414,
        0.1054027 , -0.20183711, -0.04462275, -0.10241827,  0.1500399 ,
       -0.00371991, -0.14587541, -0.13629062,  0.14532116, -0.12076889,
       -0.10334043,  0.12972057, -0.07136378, -0.14081767, -0.25441515,
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       -0.11728365, -0.15155602,  0.03145567,  0.20102592, -0.01978359,
       -0.02661396,  0.18780899,  0.10887503,  0.01717681,  0.03489362,
        0.06896196, -0.14167705, -0.04815923, -0.13087641, -0.01773774,
        0.02390907, -0.13613394,  0.04634035,  0.08272897, -0.08410635,
        0.23618044, -0.04256587,  0.00097525, -0.00881481,  0.02101083,
       -0.07304888,  0.04174875,  0.16881356, -0.31508178,  0.24252096,
        0.19326195,  0.05535217,  0.10043406,  0.13705032,  0.05888433,
        0.0262608 ,  0.05310968, -0.11336857, -0.1998698 , -0.01852458,
       -0.09978136,  0.05086243,  0.02420163])
```

```
distance: [0.48498727]
```

Similarity (<0.6 is a match)



00.jpg



01.jpg



02.jpg



thomas.jpg



03.jpg



04.jpg



05.jpg



06.jpg



07.jpg



08.jpg



09.jpg

```
00.jpg - distance: [0.42451057]
01.jpg - distance: [0.40819279]
02.jpg - distance: [0.36377479]
03.jpg - distance: [0.4821185]
04.jpg - distance: [0.50907727]
05.jpg - distance: [0.59712928]
06.jpg - distance: [0.46968764]
07.jpg - distance: [0.48498727]
08.jpg - distance: [0.58143877]
09.jpg - distance: [0.52975936]
```

Clustering of faces

- Thomas pic * 11

+



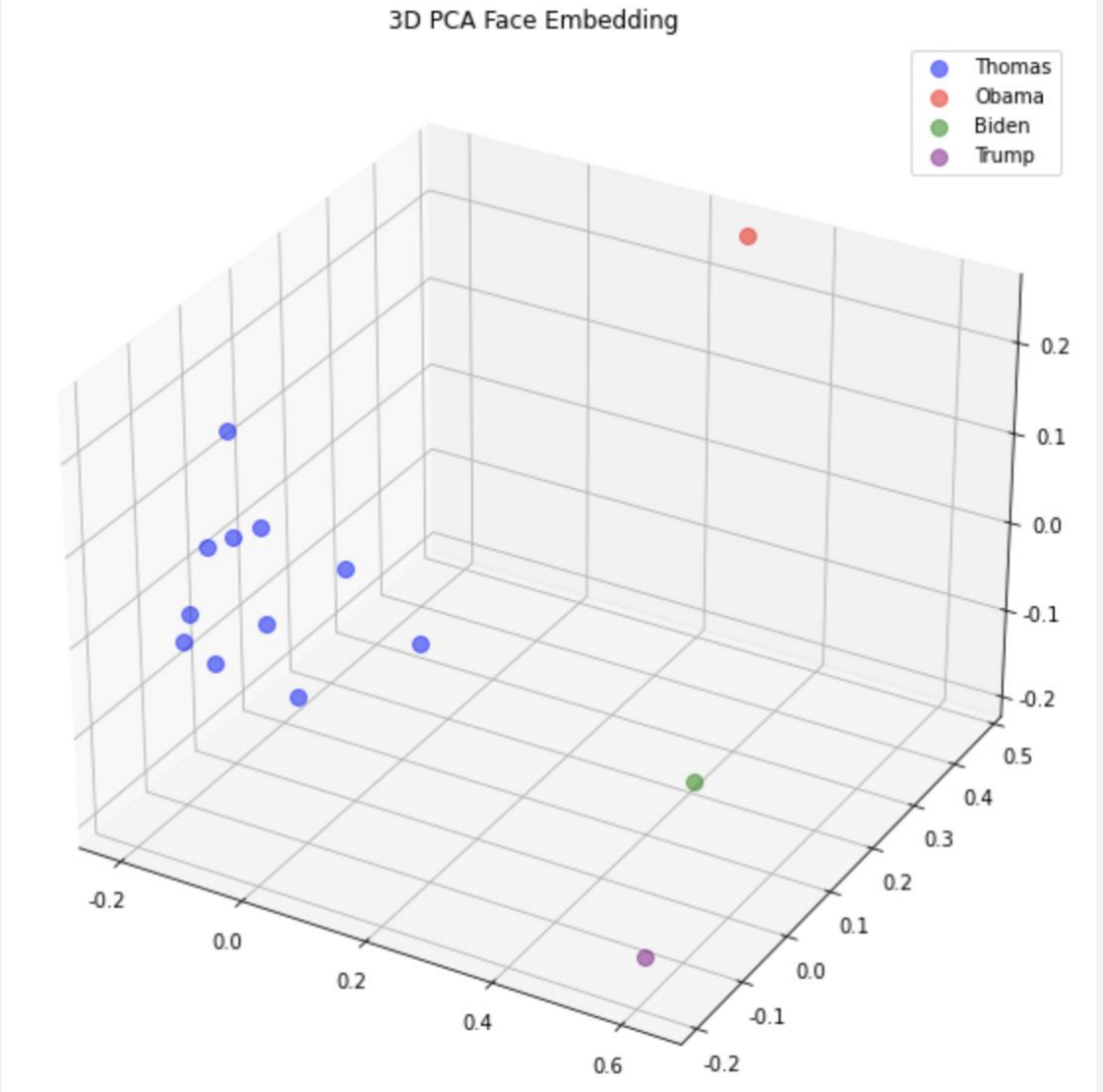
trump.jpg



biden.jpg



obama.jpg



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- **Lab (Demo): Unsupervised Learning**

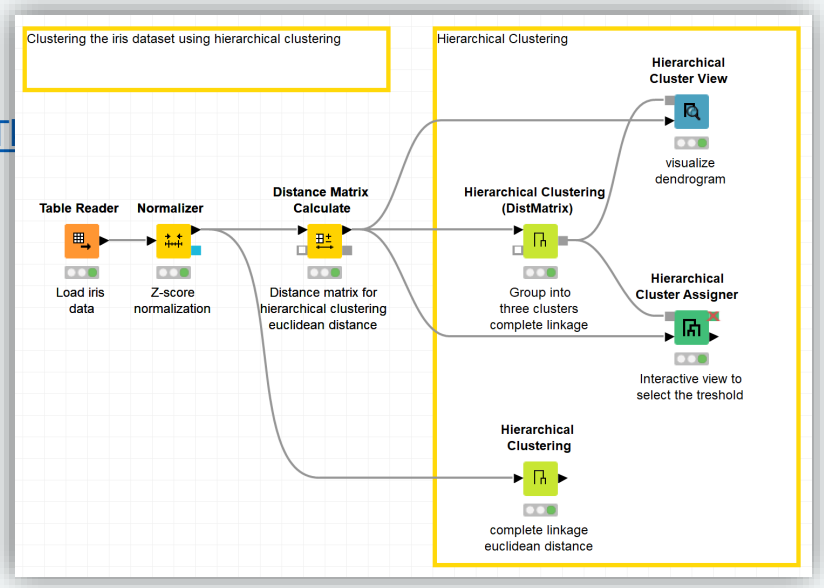
Datasets

Dataset used : iris dataset

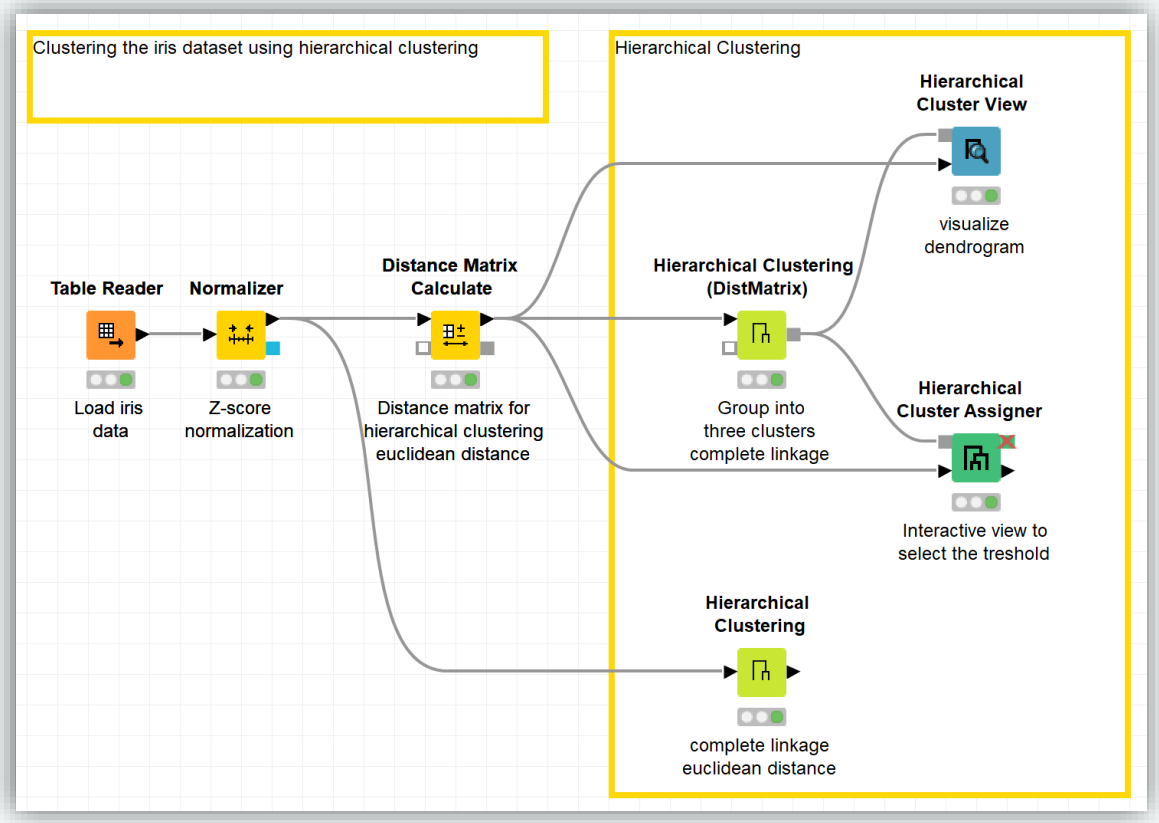
Sample Workflows:

- Hiearchical Clustering“ https://kni.me/w/rlXFxYxQmbgNgSsM
  - Normalization
  - Distance calculation
  - Hiearachical clustering

- K-means clustering“ https://kni.me/w/t8UVEQH1sTT
  - Partitioning
  - Numeric error measures

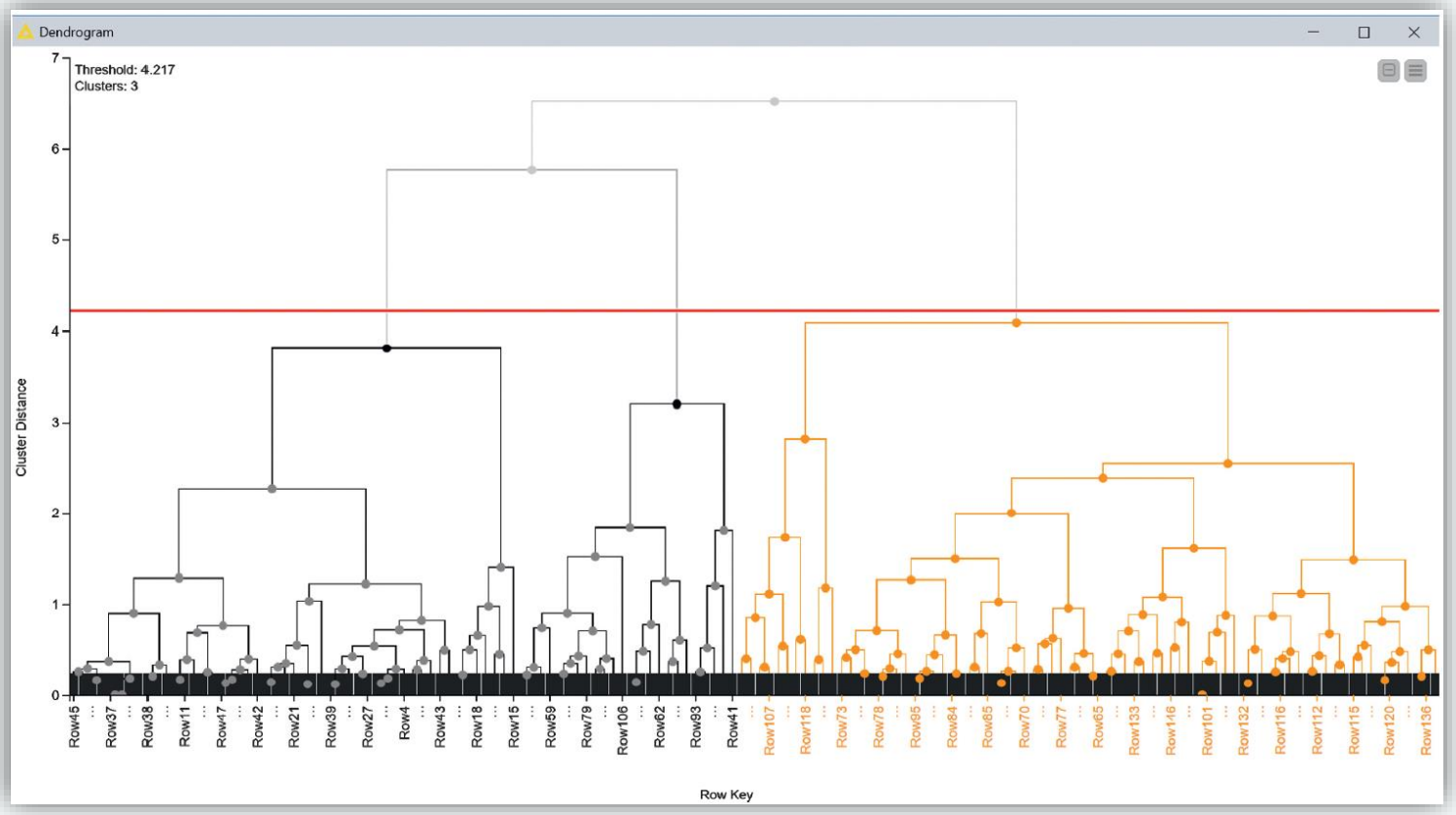


Hierarchical Clustering



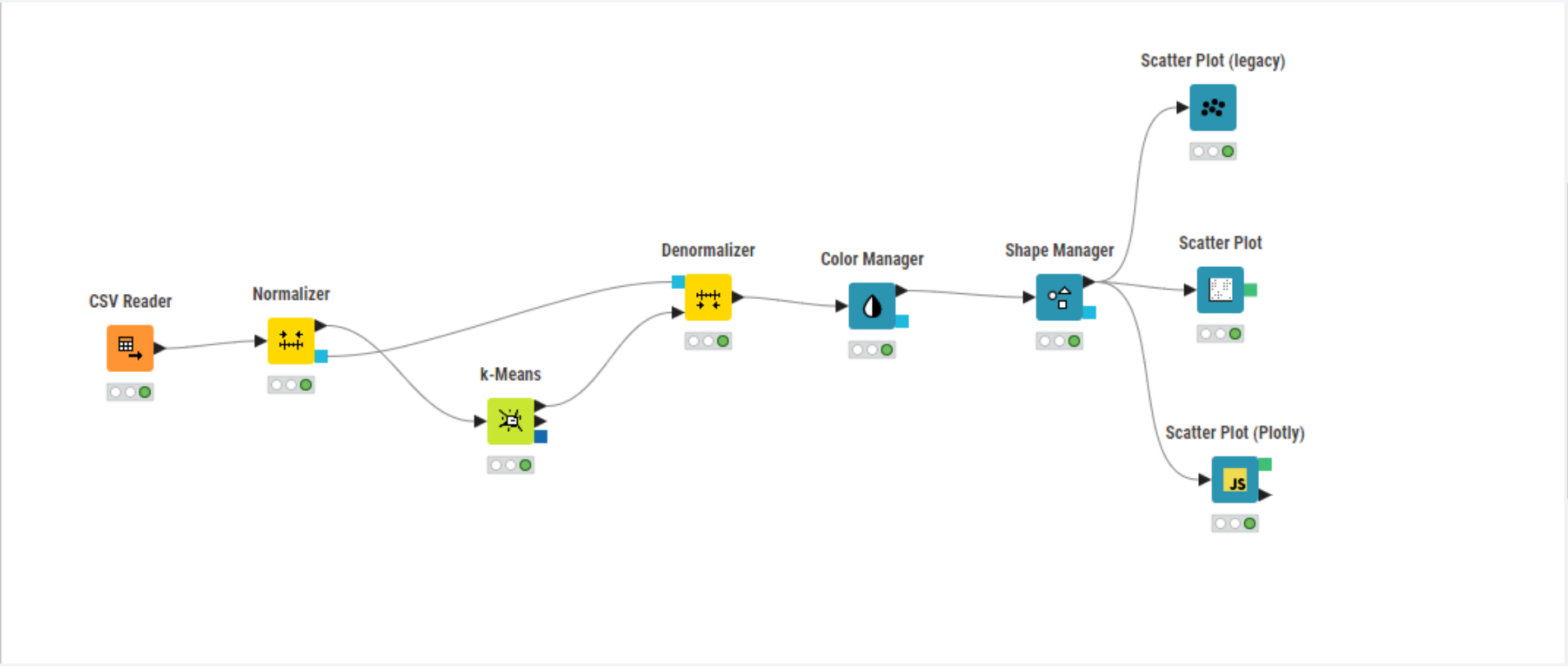
Workflow implementing hierarchical clustering with the simple Hierarchical Clustering node and with the more complex sequence of nodes, including the Distance Matrix Calculate node

Hierarchical Clustering



A dendrogram for the iris data obtained with Euclidean distance and complete linkage. Moving the threshold line changes the number of clusters and the assignment for the input rows.

K-Means Clustering



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- **Review Questions**

Review Questions

Question 1 (20 points): Explain what are supervised learning and unsupervised learning and what are their main differences.

Question 2 (20 points): Explain the differences between classification and regression with examples (you can use figures to explain their differences).

Question 3 (20 points): Explain the meaning and the components of the loss function

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - f(x_i))^2$$

Discuss, what is y_i , $f(x_i)$ and the power of 2 expression.

Review Questions

1. Which of the following can be considered unsupervised learning?

(A) Sale Forecast

(B) Find communities of internet users

(C) Predict spending on food

(D) **Determine whether market sentiment is good or bad**

2. Which one of the following is an application of “regression” (other than “classification”)?

(A) Determine whether there is a person in a picture

(B) Predict the future price of a stock

(C) Predict whether a person is married

(D) Predict whether is will rain

3. We should measure the accuracy of a classifier based on …

(A) the training set

(B) the test set

(C) the whole data set

(D) None of them

4. In a regression we can use the following measure to assess prediction performance?

(A) Gini Index

(B) Manhattan Distance

(C) R-Squared

(D) **Correlation**

Review Questions

5. In a clustering analysis we can use one of the following measure to form clusters

(A) Gini Index (B) Manhattan Distance (C) R-Squared (D) **Correlation**

6. Consider the following table for accuracy measurement, what is the “accuracy” of the model (TP+TN)/(TP+TN+FP+FN)?

	Predicted: Yes	**Predicted: No**
Actual: Yes	90	40
Actual: No	10	60

(A) 90/200 (B) 90/100 (C) 50/100 (D) 150/200